

Household Expectations in Uncertain Times*

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Abstract

How does uncertainty shape household expectations? Using U.S. survey data, we show that households revise their inflation expectations less when uncertainty stems from noisy new information, and more when it reflects uncertainty about prior beliefs. We then estimate a belief-updating model to disentangle the post-2020 surge in uncertainty. During the initial pandemic, higher macroeconomic volatility raised prior uncertainty, prompting households to seek new information. In contrast, during the high-inflation period, households perceived new information as noisier and relied more on their priors. Our findings reveal distinct effects of uncertainty sources on expectations and provide a tractable framework for separating them.

Keywords: Expectations, inflation, household surveys, consumer beliefs, information frictions, sources of uncertainty.

JEL Classification: D81, D83, D84, E31.

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The modern economy is an “uncertainty economy”. Rising uncertainty is pervasive, from earnings calls to economic commentaries, and especially evident in surveys of business and consumer expectations.¹ At the same time, the large shifts in economic expectations during the post-pandemic inflation surge drew renewed interest in how agents form beliefs and respond to new information. Yet, despite growing consensus that attention and belief formation are central to macroeconomic dynamics, economic growth, asset prices, and the transmission of monetary policy,² we know relatively little about how uncertainty shapes the way agents process new information and form expectations.

We address this gap by studying how *subjective uncertainty* influences the formation of inflation expectations in household survey data. Our key insight is that the source of uncertainty matters. When prior beliefs are more uncertain, households perceive new information as relatively more precise and update more. Conversely, when uncertainty arises from noisier new information, households update less. We use these insight to unpack the sources of the post-pandemic surge in uncertainty. We find that the initial COVID-19 outbreak was associated with higher macroeconomic volatility, which rendered existing information obsolete and prompted households to update their beliefs. In contrast, the subsequent high-inflation period was characterized by noisier information, leading households to update less. Our take-away is that different sources of uncertainty have distinct effects on macroeconomic dynamics through expectations, and that policymakers can use survey-based measures of belief updating to disentangle the underlying source of uncertainty.

Our empirical analysis draws on the U.S. Survey of Consumer Expectations (SCE), a large monthly rotating panel eliciting households inflation expectations. Two features of the SCE are crucial for our strategy. First, the survey collects not only point forecasts but also density forecasts, which allows us to measure individual subjective uncertainty (Manski, 2004, 2018). Second, its large panel structure allows us to measure “*belief stickiness*” — how much weight households place on prior information relative to new information when revising their beliefs — and track how it evolves over time and across socioeconomic groups.³

¹After the COVID-19 pandemic, the Survey of Consumer Expectations (SCE) recorded the largest and persistent rise in subjective uncertainty since the survey began over a decade prior (Armantier et al., 2021).

²See, for e.g., Van Nieuwerburgh and Veldkamp (2006); Kozłowski et al. (2020); Maćkowiak and Wiederholt (2015).

³We adopt a recent methodology based on the cross-sectional covariance between current and lagged forecasts (Goldstein, 2023), which enables us to leverage the rich cross-sectional variation to track belief updating over time and across sociodemographic groups. The more widely used strategy of Coibion and Gorodnichenko (2015a) relies on time-series variation in consensus forecasts to estimate stickiness, but it requires long time spans that are rarely available in consumer surveys. In contrast, (Goldstein, 2023)’s approach exploits cross-sectional variation in individual expectations, enabling estimation even with surveys

Our analysis proceeds in four steps. First, we document a striking reversal in the relationship between belief stickiness and uncertainty in the post-pandemic economy. During the early months of COVID-19, households updated beliefs more despite a sharp increase in uncertainty. By contrast, during the subsequent high-inflation period, uncertainty continued to rise while beliefs became stickier. This shift in the co-movement of uncertainty and belief updating raises questions about the mechanisms through which households respond to uncertainty and motivates our analysis.

Second, to interpret this evidence, we show that the source of uncertainty matters. Using the pre-pandemic sample, we build proxies for the two components of subjective uncertainty: uncertainty about prior beliefs and uncertainty about new information. We find that higher prior uncertainty reduces stickiness (i.e., increases updating), while greater noise in new information raises stickiness. The effects are sizable: a one standard deviation increase in prior uncertainty lowers belief stickiness by approximately 20%, while a similar increase in new information noise raises it by about 30%. These patterns are *qualitatively* consistent with Bayesian updating, but not quantitatively. In particular, our empirical estimates of belief stickiness are on average lower than those implied by the Bayesian benchmark, indicating that households overreact to new information. To quantify this bias, we estimate a belief-updating model through moment matching, which delivers an empirically disciplined belief-formation rule with belief stickiness as a function of prior uncertainty and information noise.

Third, we use this empirically estimated framework to decompose the sources of aggregate uncertainty in the post-pandemic period. We show that the initial surge in uncertainty at the COVID outbreak reflected higher prior uncertainty, pointing to a potential regime shift in macroeconomic volatility or policy uncertainty. This structural change rendered existing information obsolete, encouraging households to seek out new information to update their beliefs. In contrast, during the following high-inflation period, belief stickiness increased alongside rising inflation uncertainty, suggesting a relative increase in the noise of new information. Faced with less reliable signals, households relied more on their prior beliefs.

Fourth, we investigate the role of lockdown policies in driving belief dynamic at the onset of the COVID-19 pandemic. We show that variation in the intensity of state-level lockdowns and are associated with lower belief stickiness and lower subjective uncertainty. These findings suggests that mobility restrictions and the widespread shift to remote work may have lowered the marginal cost of acquiring information, allowing households to access

that have shorter time-series dimensions.

more accurate signals. Yet, lockdowns cannot account for the *concurrent* surge in overall uncertainty at the pandemic onset, supporting the evidence of an increase in macroeconomic volatility.

From a policy perspective, our findings have two key implications. First, belief stickiness can help identify the underlying source of uncertainty. As high-frequency expectations data become increasingly available, this offers a useful diagnostic tool for developing policies to address uncertainty: if uncertainty reflects noisier signals or higher information costs, communication policies can mitigate its effects, whereas if it reflects increased fundamental volatility, communication strategies alone may have a more limited impact. Second, we contribute to the understanding of household belief formation under uncertainty. As uncertainty becomes a defining feature of the economic environment, its impact on belief dynamics is critical for policy. When uncertainty is driven by noisy information, households exhibit greater belief stickiness, which adds inertia to macroeconomic adjustment (Maćkowiak and Wiederholt, 2015), weakens the effectiveness of central bank communication such as forward guidance (Angeletos and Lian, 2018), and flattens the Phillips curve (Afrouzi and Yang, 2021). Conversely, when uncertainty reflects fundamental volatility, belief stickiness is lower, resulting in faster belief adjustment and markedly different implications for policy transmission.

Contributions to the literature On the empirical side, we add to the literature measuring information frictions in expectation surveys (Mankiw and Reis, 2002; Coibion and Gorodnichenko, 2015a; Kohlhas and Walther, 2021; Benhima and Bolliger, 2022; Gemmi and Valchev, 2023). Unlike most of these studies, which focus on professional forecasters, we focus on household survey data. We build on the empirical strategy of Goldstein (2023), who documents a decline in belief stickiness among professional forecasters in the early stages of the COVID-19 pandemic, but finds no such pattern in the Michigan Survey of Consumers. Pfäuti (2023) also measures belief stickiness in the Michigan Survey of Consumers, but using the consensus forecast, and relates it to the level of realized inflation. Relative to this work, we leverage the higher frequency and panel structure of the SCE to identify the time variation in household belief stickiness. More importantly, we estimate how belief updating varies with individual-level subjective uncertainty.⁴

⁴De Bruin et al. (2011) also examines subjective uncertainty in the SCE and documents that more uncertain consumers revise their beliefs more. Instead, we consider both posterior and prior uncertainty and estimate their impact on information rigidities. Baker et al. (2020) proxies uncertainty shocks with natural disasters and finds that they decrease both belief stickiness and forecast errors in the Consensus Economics Surveys. Instead, we directly measure individual subjective uncertainty and show that the relation with

We also contribute to a large empirical and theoretical literature on the measurement and consequences of macroeconomic uncertainty (Bloom, 2009; Jurado et al., 2015; Baker et al., 2016; Bloom et al., 2018; Baley and Blanco, 2019), especially the ones measuring uncertainty with survey data (Manski, 2018; Binder et al., 2022; Kumar et al., 2023; Fermand et al., 2024; Wang, 2024; Coibion et al., 2024; Georgarakos et al., 2024; Binder et al., 2025). A common limitation in this literature is that existing measures often conflate distinct sources of uncertainty—such as macroeconomic volatility and information noise—which are conceptually and statistically different (Kozeniauskas et al., 2018). We show that belief stickiness, measured in survey data, is a useful statistic for disentangling these sources: while both increase subjective uncertainty, they have opposite effects on updating. Gambetti et al. (2023) pursue a related goal using forecast disagreement to separate volatility from noise, under the assumption that the former reduces disagreement while the latter increases it. While in standard Bayesian models the relationship between disagreement and these sources depends on model calibration, the effect of each source on belief stickiness is theoretically unambiguous, making it a particularly robust indicator for this purpose.

Third, we contribute to the literature testing the rational expectation (RE) hypothesis (Bordalo et al., 2020; Broer and Kohlhas, 2024; Baley and Turén, 2023; Adam et al., 2025). This literature documents systematic overreaction to new information, identified through the predictability of individual forecast errors. These tests generally require tracking the same individuals over long periods, which has confined most of the empirical work to surveys of professional forecasters. In contrast, we test the RE hypothesis by examining the relationship between belief stickiness and uncertainty implied by the Bayesian updating framework. This approach allows us to study deviations from rationality using shorter panel data with richer cross-sectional variation—such as consumer expectation surveys. We document a clear overreaction bias, particularly among low-numeracy and younger households. Our contribution to this literature is threefold. First, consumer expectations often differ substantially from those of professional forecasters, making it important to study them separately. Second, our approach enables the analysis of how belief distortions vary across socioeconomic characteristics. Third, RE tests based on professional forecasters may conflate deviations from rationality with strategic diversification or reputational incentives (Gemmi and Valchev, 2023).⁵

A growing body of research applies randomized controlled trials (RCTs) to study how belief stickiness depends on the source of uncertainty.

⁵Overreaction has also been documented on analyst forecasts (Bordalo et al., 2019, 2024), market reactions to economic news (Bianchi et al., 2025), and laboratory experiments (Afrouzi et al., 2023; Ba et al., 2024).

information treatments shape expectations (Armantier et al., 2016; Cavallo et al., 2017; Armona et al., 2019; Roth and Wohlfart, 2020; Coibion et al., 2022; Link et al., 2023). A common finding is that agents update their beliefs in line with Bayesian rationality, responding more when prior beliefs are more uncertain and when signals are more informative.⁶ Rather than relying on experimental variation, we leverage *naturally occurring* variation in prior uncertainty and belief updating, which allows us to examine the time-variation in belief stickiness and address the external validity concerns inherent in RCTs. Our approach is complementary to recent findings by Weber et al. (2025), who show that information treatments become less effective during periods of high inflation, potentially due to increased pre-treatment attention which lower prior uncertainty (Mackowiak and Wiederholt, 2024). We directly test this mechanism by examining how belief adjustment varies with prior uncertainty, and find that households update less when they perceive their prior information to be more accurate.⁷

Roadmap The rest of the paper is organized as follows. Section 1 presents the general framework that guides our empirical strategy, which we implement in Section 2. Section 3 studies the mechanisms underlying the relation between belief stickiness and uncertainty. Section 4 investigates the impact of lockdowns on belief stickiness. Section 5 concludes.

1 A general framework of belief updating

We present a general theoretical framework embedding different models of belief updating, which will guide our empirical strategy. Households believe that inflation x_t following an AR(1) process

$$x_t = (1 - \rho)\mu_x + \rho x_{t-1} + u_t, \quad (1)$$

where $u_t \sim N(0, \sigma_u^2)$ is the fundamental noise and μ_x the long-run mean.

Households form beliefs about the unobserved fundamental by observing a private signal, the accuracy of which we allow to differ across groups of consumers. Formally, in time t ,

⁶This pattern is well-documented for inflation expectations. For other variables, however, the evidence is mixed: Fuster et al. (2022) find the opposite effect for housing price expectations, while Armona et al. (2019) and Conlon et al. (2018) find no significant effects of uncertainty in the housing and labor markets.

⁷A related literature studies endogenous information acquisition and attention allocation by consumers and firms (Roth et al., 2022; Mikosch et al., 2024; Link et al., 2024). Rather than analyzing the determinants of attention choices, we measure the outcome of that process—the quantity of information held—through subjective uncertainty. As we show in Section 3, in the Bayesian framework, belief stickiness depends solely on this quantity, regardless of its source.

household i in group j forms beliefs about the variable realization at horizon $t + h$ after observing a private signal⁸

$$s_{i,j,t} = x_{t+h} + e_{i,j,t}, \quad (2)$$

where the signal noise $e_{i,j,t} = \eta_{i,j,t} + \omega_t$ consists of two components. first, an idiosyncratic component $\eta_{i,j,t} \sim N(0, \sigma_{\eta,j,t}^2)$ i.i.d. across time and households, i.e., $\int^i \eta_{i,j,t} di = 0$, but with similar variance across households in group j ; second, a common component $\omega_t \sim N(0, \sigma_{\omega,t}^2)$ i.i.d. only across time, but correlated across agents. Let $\sigma_{e,j,t}^2 \equiv \sigma_{\eta,j,t}^2 + \sigma_{\omega,t}^2$ define the overall variance of the signal noise.⁹

Each household i in group j forms beliefs $E_{i,j,t}[x_{t+h}]$ at time t about the variable at h periods ahead according to

$$\begin{aligned} E_{i,j,t}[x_{t+h}] &= (1 - G_{j,t})E_{i,j,t-1}[x_{t+h}] + G_{j,t}s_{i,j,t} \\ E_{i,j,t-1}[x_{t+h}] &= (1 - \rho)\mu_x + \rho E_{i,j,t-1}[x_{t+h-1}], \end{aligned} \quad (3)$$

where $G_{j,t}$ is the weight households assign to new information and $E_{i,j,t}[x_{t+h}]$ is a potentially non-rational expectation operator, conditional on the information set of agents i in group j at time t about x_{t+h} . We refer to $G_{j,t}$ as “gain” and $1 - G_{j,t}$ as “stickiness”.

We make no assumption about what determines the weight on new information G_t . At this stage, we only require that posterior beliefs are a linear combination of the prior and the signal. This general framework encompasses a wide range of belief-updating models, including the rational Bayesian case, where G_t corresponds to the Kalman gain.¹⁰ The framework also embeds several behavioral models, such as Diagnostic Expectations (Bordalo et al., 2020) and overconfidence (Broer and Kohlhas, 2024), as detailed in Appendix A.

Based on equation (3), we derive the second moment of the individual subjective distribution, i.e., the individual uncertainty, defined as $\Sigma_{i,j,t+h,t} \equiv \text{var}_{i,j,t}(x_{t+h} - E_{i,j,t}[x_{t+h}])$:

$$\sigma_{i,j,t+h,t}^2 = (1 - G_{j,t})^2 \sigma_{i,j,t+h,t-1}^2 + G_{j,t}^2 \sigma_{e,j,t}^2, \quad (4)$$

⁸Our assumption of an unbiased signal centered on the realization is without loss of generality under mild conditions. If agents instead observe a more general signal $\tilde{s}_{i,j,t} = \alpha_1 + \alpha_2 x_{t+h} + \phi_{i,j,t}$ with $\phi_{i,j,t} \sim N(0, \sigma_{\phi,j,t}^2)$, they can extract information by transforming it as $s_{i,j,t} \equiv \tilde{s}_{i,j,t} - \alpha_1 / \alpha_2 = x_{t+h} + e_{i,j,t}$, where $e_{i,j,t} \equiv \alpha_2^{-1} \phi_{i,j,t}$. This corresponds to our specification.

⁹The assumption of normality of the signal error term is consequential for our analysis and shared by the existing literature, e.g. Coibion and Gorodnichenko (2015b); Bordalo et al. (2020).

¹⁰The weight on new information in the Bayesian rational expectation case is the Kalman gain, which equals $G_{j,t}^{RE} = \frac{\sigma_{j,t+h,t-1}^2}{\sigma_{e,j,t}^2 + \sigma_{j,t+h,t-1}^2}$ where $\Sigma_{j,t+h,t-1} = \text{var}_t(\mathbb{E}_{i,j,t-1}[x_{t+h}] - x_{t+h})$ is the prior variance.

where $\sigma_{i,j,t+h,t}^2 \equiv \text{var}_{i,j,t}(x_{t+h} - E_{i,j,t-1}[x_{t+h}])$ is the prior uncertainty and $\sigma_{e,j,t}^2$ the new information noise. Equation (4) implies that, as long as information quality and belief stickiness is the same across households in group j , then belief uncertainty is also the same: $\sigma_{i,j,t+h,t} = \sigma_{j,t+h,t}$.¹¹

Before proceeding, we clarify some of the terminology used throughout the paper. We refer to $1 - G_t$ as belief stickiness, or information rigidity (Coibion and Gorodnichenko, 2012, 2015a). Another widely used term in the recent literature is “inattention” (Goldstein, 2023; Pfäuti, 2023; Weber et al., 2025). However, this term may be confused with its use in the endogenous information literature, where attention refers to the costly acquisition of information that reduces the noise in new signals $\sigma_{e,t}$ (Mikosch et al., 2024; Link et al., 2024). While in a rational setting the two concepts are strictly related, we remain agnostic about household rationality and therefore prefer a more general terminology. Another related term is belief *anchoring*, which refers to long-run inflation expectations being stable and closely aligned with a central bank’s inflation target. The literature requires long-term horizons, typically 5 to 10 years, to identify beliefs about the central banks’ target or the steady state inflation level (for example, Gürkaynak et al., 2005; Carvalho et al., 2023). Instead, the 3-year horizon we consider is too short to relate to the inflation target directly.

2 Households’ belief stickiness

2.1 Data

We measure realized inflation using the monthly “Consumer Price Index for All Urban Consumers” (CPIAUCSL) from the FRED database. To align with the survey question, we convert it into a year-over-year growth rate.

Expectation data comes from the *Survey of Consumer Expectations* (SCE), a monthly survey of a rotating panel of approximately 1,200 household heads collected by the Federal Reserve Bank of New York (FRBNY) since late 2012.¹² Participants remain in the panel for up to 12 months, with a roughly equal number entering and exiting each month. We use the core survey sample from June 2013 to May 2023, which includes monthly point and density forecasts of inflation, along with socioeconomic and background information. The dataset

¹¹Since in this framework, as in a large set of belief formation models, the source of variation in uncertainty is the same as in belief stickiness, it is not possible to estimate G_t by regressing posterior and prior uncertainty with simple OLS.

¹²The respondents are household heads, defined as “the person in the household who owns, is buying, or rents the home”. See Armantier et al. (2017) for additional details.

covers 108 months, with about 1,300 observations per month—roughly 130,000 respondent-month observations from around 20,000 unique individuals. We consider point forecasts only if respondents provide a meaningful density forecast (i.e. the survey provides the variance) and if the point forecast is contained in the support of the density forecast. Moreover, in each month, we drop the point forecast observations for each horizon at the top and bottom 1 percentiles to avoid outliers.

The SCE asks respondents to provide expectations about future inflation at two different horizons: expected inflation/deflation over the next 12 months (which we define as “1 year”) and expected inflation/deflation over the 12 months starting from 24 months in the future (which we define as “3 years”). The SCE asks respondents to indicate both their point forecast for future expected inflation and their subjective distribution over all possible inflation realizations. We focus on the 3-year horizon and use the shorter horizon forecasts for robustness in Appendix N.

Expectation mean First, we use point forecasts to measure the first moment of the individual subjective expectations—that is, expected inflation. We measure the posterior expectations $E_t^i[x_{t+h}]$ as the point forecast provided in month t about the horizon h . To construct prior expectations $E_{t-1}^i[x_{t+h}]$, we rely on the structure imposed in equation (1), using the lagged posterior mean $E_{t-1}^i[x_{t+h-1}]$. In Appendix D, we estimate an autoregressive process for yearly inflation across different subsamples (see Table A.1). Our benchmark specification uses the estimated parameters $\hat{\rho} = 0.946$ and $\hat{\mu}_x = 2.3$ to construct prior expectations. In Appendix P we replicate our main results using alternative calibrations, with virtually the same results.

Expectation uncertainty Second, we use subjective density forecasts to measure posterior uncertainty.¹³ In the SCE, respondents assign probabilities to 10 symmetric bins covering a range of inflation outcomes from -12% to 12% , in increments of 2 to 4 percentage points (see Appendix B). The FRBNY constructs individual-level forecast variances by fitting parametric distributions to these responses, using the method developed by Engelberg et al. (2009) and detailed in Armantier et al. (2017).¹⁴ We refer to the standard deviation

¹³While cross-sectional belief *dispersion* is sometimes used to measure *uncertainty*, estimating it from subjective density forecasts provides two advantages: first, we can measure uncertainty at the individual household level; second, the theoretical relationship between information noise and belief dispersion is not the same as with belief uncertainty, as we show in Appendix L.

¹⁴A potential concern is that the upper limit of the survey bins may be too low during periods of high inflation. In such cases, clustering of responses in the top bin could lead to measurement error in the

of the subjective distribution reported in month t for horizon h as posterior uncertainty, denoted $\sigma_{i,t+h,t}$.

As described above, we use point forecasts to capture the first moment of subjective beliefs and density forecasts to measure the second moment. This separation avoids mechanical correlations between the two, such as those arising from measurement error. Nonetheless, in Appendix O, we show that our main results are robust to using the mean of the density forecast as a measure of expected inflation.

Sociodemographics For each respondents, the survey reports gender, age and race. Moreover, we construct an indicator variable with value one if the respondent attended college and zero otherwise. The survey provides respondent income, but only as a categorical variable. We construct an indicator with value 1 if the respondent has an income lower than 50k, between 50k and 100k, and above 100k. The SCE also reports respondents’ numeracy, based on their ability to answer questions about probabilities and compound interest (Lusardi, 2008). Respondents who answer at least four out of the five questions correctly are assigned a high numeracy indicator. Finally, tenure indicates the number of months each respondent has been present in the survey, from 1 to 12. Kim and Binder (2023) shows that tenure in the survey has crucial effects on inflation expectations and uncertainty, by prompting information acquisition between survey waves, i.e. “learning-through-survey”. Therefore, we include tenure as a determinant of belief stickiness.

2.2 Estimating belief stickiness

We estimate belief stickiness by comparing individual posterior with prior forecasts across households, building on the recent methodology developed in Goldstein (2023) and Gemmi and Valchev (2023). In contrast, earlier studies rely on the approach introduced by Coibion and Gorodnichenko (2015a), which estimates belief stickiness by regressing consensus forecast errors on forecast revisions. However, this method is not well suited to our setting for two main reasons. First, as noted in their Appendix, it yields biased estimates in the presence of common noise in the signal structure ($\sigma_\omega > 0$ in our framework). Second, it requires a long time-series dimension that is unavailable in the SCE. The methodology we adopt addresses both limitations by leveraging cross-sectional variation in individual forecasts.

uncertainty measure. However, in Appendix C we show that our bin-based uncertainty measure closely tracks an alternative based instead point forecasts rounding, as in Binder (2017).

Demeaning (3) using consensus forecasts, we obtain¹⁵

$$E_{i,j,t}[x_{t+h}] - \bar{E}_{i,j,t}[x_{t+h}] = (1 - G_{j,t})(E_{i,j,t-1}[x_{t+h}] - \bar{E}_{j,t-1}[x_{t+h}]) + G_{j,t}\eta_{i,j,t-1} \quad (5)$$

where $\bar{E}_{j,t-1}[x_t] = \int^{i \in j} E_{i,j,t-1}[x_{t+h}] di$ is the average forecast in group j at time t . Equation (5) provides an unbiased strategy to measure information rigidity at t for each group j :

$$E_{i,j,t}[x_{t+h}] = \alpha_{j,t} + \beta_{j,t}E_{i,j,t-1}[x_{t+h}] + err_{i,j,t} \quad (6)$$

where i indicates the household and t the year-month. The coefficient $\beta_{j,t}$ is an unbiased estimator of the belief stickiness $1 - G_{j,t}$. Intuitively, higher belief stickiness implies a higher correlation between posterior beliefs and prior beliefs (higher $\beta_{j,t}$), while lower belief stickiness implies a lower correlation between posterior beliefs and prior beliefs (lower $\beta_{j,t}$). Our belief stickiness measure, $\beta_{j,t}$, captures the average responsiveness of household expectations to new information, incorporating both the extensive margin (whether consumers update at all) and the intensive margin (how much they update, conditional on updating).

2.2.1 Heterogeneity in belief updating

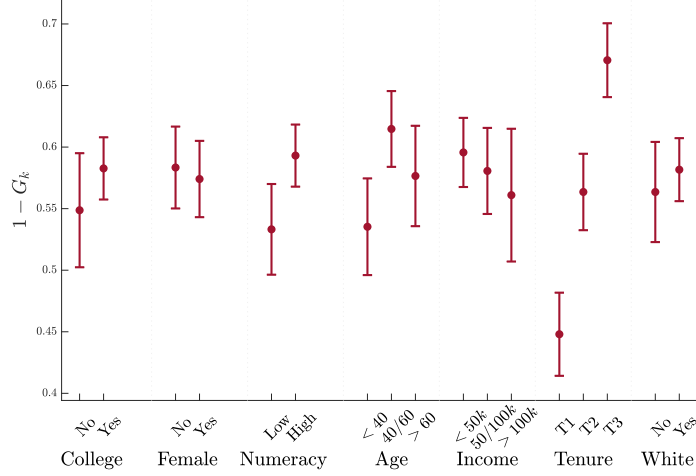
Estimating equation (6) requires selecting the socioeconomic characteristics used to group households by similarity in information quality and belief stickiness. In this Section we abstract from time variation in belief updating and instead investigate how belief stickiness, $1 - G_j$, varies across observable household characteristics. Because we abstract from time-varying stickiness, we can use group dummies interactions to demean posterior expectations and study group-specific belief stickiness:

$$E_t^i[x_{t+h}] = \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \mathbf{X}_{i,t} \mathbf{B}_2 + E_{t-1}^i[x_{t+h}] \times \mathbf{X}_{i,t} \mathbf{B}_3 + \gamma_t + \beta_4(\gamma_t \times \mathbf{X}_{i,t}) + \epsilon_t^i, \quad (7)$$

where $\mathbf{X}_{i,t}$ is a vector containing a set of socioeconomic characteristics binary indicators $X_{i,k,t}$, and $\mathbf{B}_3$ is a vector of coefficients capturing their impact on belief stickiness. The characteristics we consider are the following: tercile of tenure (i.e. number of months in the survey), whether they hold a college degree, whether their age is over 60 or under 40, income over 100k or below 50k, high numeracy, gender, and race. The average belief rigidity

¹⁵Demeaning the belief updating equation eliminates the actual realization of the underlying process, which could represent only part of the actual variable realization observable by the econometrician. In other words, the econometrician does not need to observe x_t to estimate belief stickiness.

Figure 1: Belief updating across groups



Notes: The figure reports the average marginal effects of $E_{t-1}^i[x_{t+h}]$ for each socioeconomic characteristic k in (7), which identifies the average belief stickiness associated with that characteristic:

$1 - \hat{G}_k = \hat{\beta}_1 + \hat{\beta}_{3,k} + \sum_{k' \neq k} \hat{\beta}_{3,k'} \bar{X}_{i,k',t}$, where $\bar{X}_{i,k',t} = \frac{1}{N} \sum_{i,t} X_{i,k',t}$. Table A.3 reports the detailed results. Sample: 2013M6-2023M5. Standard errors are clustered at the household and time level. Bars represent the 95% confidence interval.

of socioeconomic characteristic k is identified by the average marginal effect of $E_{t-1}^i[x_{t+h}]$ associated with characteristic k , i.e. $1 - G_k = \beta_1 + \beta_{3,k} + \sum_{k' \neq k} \beta_{3,k'} \mathbb{E}[X_{i,k',t}]$.

We find that belief updating differs significantly across socioeconomic groups. Figure 1 plots belief stickiness by characteristic k (see detailed regression results in Table A.3). Households with longer tenure, a college degree, and higher numeracy exhibit significantly greater belief stickiness. In contrast, young respondents display lower belief stickiness.¹⁶

In the Bayesian framework, belief stickiness reflects the perceived precision of incoming information. Interpreted this way, our findings would imply that younger, less experienced, or less skilled individuals receive more accurate signals. Alternatively, households may depart from Bayesian updating, and the observed heterogeneity may reflect behavioral biases rather than differences in information quality. In the next section, we investigate this distinction by testing the Bayesian framework using direct measures of perceived information accuracy.

Robustness A potential concern with our estimate of belief stickiness is that it may be driven by inexperienced consumers not understanding or paying attention to the survey questions. However, we find the opposite: belief stickiness is significantly higher among respondents with longer tenure in the survey and higher levels of numeracy. Another concern

¹⁶For a detailed analysis of attention heterogeneity in consumer and professional forecasters surveys, see Boccanfuso and Neri (2024).

is that respondents may become fatigued over time and begin to mechanically repeat their previous answers. To address this, Figure A.2 replicates our analysis using only consumers who revise their forecasts between months t and $t - 1$. The results remain unchanged, with belief stickiness still increasing in tenure, suggesting that respondent fatigue does not substantially drive the pattern.¹⁷

2.2.2 Time variation in belief updating

We now allow belief stickiness to vary over time and examine its fluctuations around the pandemic period, as well as its correlation with belief uncertainty.

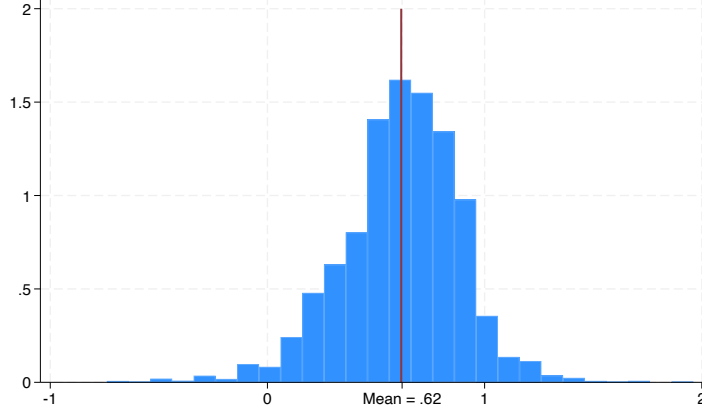
To do so, we classify consumers into groups based on the four socioeconomic characteristics that have the strongest impact on belief stickiness, as shown in Figure 1: tenure (in terciles), numeracy (high vs. low), education (college degree), and age (under 40). Each combination of these indicators defines a group, yielding a total of $J = 24$ distinct groups. We exclude group-month cells with fewer than 15 observations. We estimate regression (6) separately for each group and month, obtaining a time-series of belief stickiness estimates $\hat{\beta}_{j,t} = 1 - G_{j,t}$. Figure 2 displays the distribution of these estimates. For each month t , we compute aggregate belief stickiness as the sample-size-weighted average across groups.

We find an average belief stickiness of $\overline{1 - G} = 0.62$, which implies a new information gain of $\overline{G} = 0.38$ —that is, households place more weight on prior beliefs than on new information when forming beliefs. This level of rigidity is higher than estimates from surveys of professional forecasters (Coibion and Gorodnichenko, 2015a; Goldstein, 2023), but in line with strategic incentives-corrected estimates in Gemmi and Valchev (2023). This may reflect households’ lower attention to new inflation-related news compared to professionals, or it may be driven by survey frequency: quarterly in professional forecasts versus monthly in the SCE. We note that our empirical approach does not speak to the optimality of belief stickiness—that is, whether the weight on new information G equals the Bayesian Kalman gain.¹⁸ We return to the question of rationality in Section 3.1.3. Before that, we show that

¹⁷Figure A.17 confirms the robustness of our findings to an alternative calibration of prior expectations. In Figure A.11, we replicate the analysis using 1-year-ahead inflation expectations, and in Figure A.14, we measure expectations using the mean of consumers’ density forecasts rather than point forecasts. In both cases, we find similar patterns for tenure, though the relationship with numeracy is weaker.

¹⁸Some estimates of belief stickiness fall outside the rational expectations (RE) interval: approximately 3% of them are below zero, and 4% exceed one. While this may reflect measurement error, it could also stem from deviations from rational expectations. For instance, in the diagnostic expectations model of Bordalo et al. (2020), $G_t^j = (1 + \theta)G_t^{REj}$, where $\theta > 0$ is a parameter governing overreaction to new information. In that case, our estimates becomes $\hat{\beta}_t^j = 1 - (1 + \theta)G_t^{REj}$, which can fall below zero for sufficiently high θ and

Figure 2: Distribution of estimated belief stickiness



Notes: The figure plots the estimate of (6), weighted by the group-time size. The vertical line reports the weighted average. Sample period: 2013M1 - 2023M5.

this average masks substantial fluctuations in belief stickiness over time.

Belief updating around the pandemic We document a sharp decline in belief stickiness during the COVID-19 outbreak, indicating that households began incorporating more new information into their expectations. Figure 3b shows the monthly average belief stickiness across subgroups. Prior to the pandemic, stickiness averaged around 0.65, but it dropped abruptly to approximately 0.3 during the early COVID period. It then reverted to pre-pandemic levels and stabilized at a similar value throughout the high-inflation period. These findings are robust across different empirical strategies. First, Figure A.12 confirms that the same pattern holds for the shorter one-year forecast horizon. Second, Figure A.6 replicates the exercise using the full sample (rather than subgroups), yielding similar results.^{19 20}

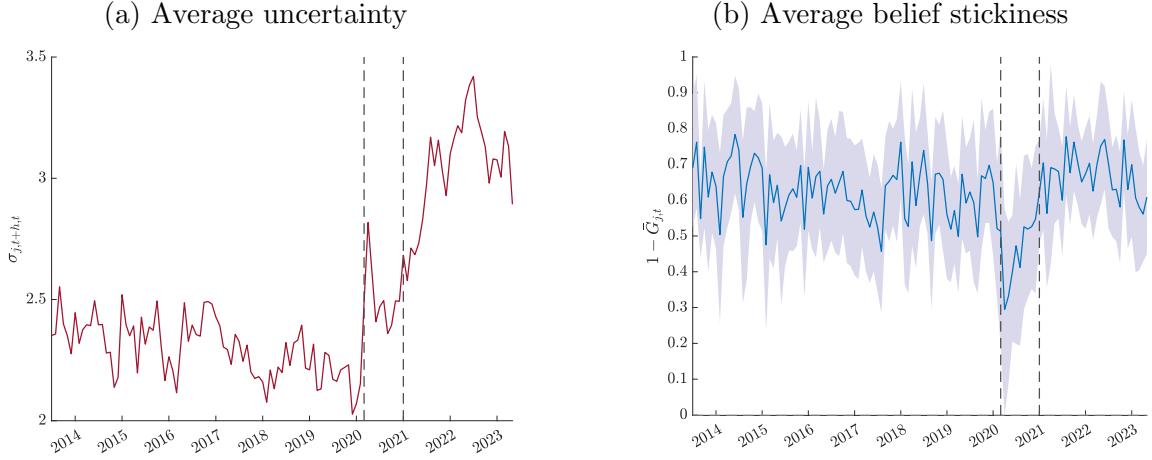
Importantly, we also document a reversal in the correlation between posterior uncertainty and belief stickiness after the onset of COVID-19, as shown in Figures 3a and 3b. During the

G_t^{REj} . Conversely, underreaction $\theta < 0$ combined with low G_t^{REj} could yield estimates above one.

¹⁹Estimating belief stickiness on the whole sample presents an advantage in terms of efficiency by increasing the sample size, but it could introduce a bias if the underlying belief stickiness actually differs across socioeconomic groups. Further details in Appendix H.

²⁰One potential concern is that changes in our estimates of belief stickiness might reflect structural shifts in the inflation autoregressive parameter ρ , which we use to construct prior expectations. We address this in two ways. First, Figure A.9 shows a sharp decline in the share of respondents in the Michigan Survey of Consumers who report not having heard any news about business conditions. This measure is independent of ρ and supports our interpretation of increased attention. Second, Figure A.8 shows no evidence of significant time variation in ρ around the pandemic period. As additional context, note that inflation itself did not begin rising until several months after the outbreak, only accelerating after January 2021.

Figure 3: Belief stickiness pre- and post-pandemic



Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 6, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across (j,t) groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

early pandemic, uncertainty surged while belief stickiness declined: consumers abandoned their priors in favor of new information, but became more uncertain about their new beliefs. In contrast, during the high-inflation period beginning in early 2021, both belief stickiness and uncertainty increased: consumers relied more on their prior beliefs while becoming increasingly uncertain.

These patterns stand in contrast to recent work that emphasizes the role of inflation levels in shaping household attention dynamics (Pfäuti, 2023; Weber et al., 2023). The large panel structure of the SCE, however, enables us to estimate belief stickiness at a monthly frequency, revealing a sharp decline already at the onset of the COVID-19 outbreak—well before inflation began to rise. Supporting this interpretation, Figure A.9 shows a drop in the share of respondents in the Michigan Survey of Consumers who report not having heard any news about business conditions. This suggests a broad increase in consumer attention to economic conditions during the early pandemic, consistent with our findings.

Such large fluctuations in belief updating stickiness may have significant macroeconomic consequences, with implications for the Phillips curve, the effectiveness of central bank communication, and the transmission of economic shocks (Afrouzi and Yang, 2021; Angeletos and Lian, 2018; Maćkowiak and Wiederholt, 2015). In this paper, we focus on understanding the drivers of fluctuations in belief updating and their relation to belief uncertainty.

Taken together, these patterns suggest that the relationship between uncertainty and belief updating may not depend solely on the level of uncertainty, but also on its source—a possibility we investigate in the next section.

3 Belief updating and uncertainty

3.1 Belief formation in the pre-COVID economy

In this section, we investigate the relationship between belief updating and belief uncertainty. First, we focus on the pre-COVID period and investigate how belief updating varies across households with different levels of prior and new information uncertainty. Second, we use these findings to structurally recover the average subjective prior and new information uncertainty and assess whether they can account for the observed decline in belief stickiness during the post-COVID period.

3.1.1 The Bayesian framework

Our empirical strategy does not require a specific belief formation model for belief stickiness $1 - G_t$ (see Section 1). However, the framework nests the noisy information rational expectations (RE) model as a special case. This allows us to assess whether the observed properties of belief stickiness are consistent with standard Bayesian updating.

Assume that households update beliefs according to Bayes' rule, receiving a noisy signal as in equation (2). Then, the posterior mean follows equation (3), and posterior uncertainty follows (4), where the belief stickiness is given by:

$$1 - G_{j,t}^{RE} = \frac{\sigma_{e,j,t}^2}{\sigma_{e,j,t}^2 + \sigma_{j,t+h,t-1}^2} \quad (8)$$

The RE model (8) yields two key testable implications. First, higher new information noise $\sigma_{e,t}^2$ is associated with higher belief stickiness. For example, households may face a higher cost of collecting information or a lower supply of information from newspapers, television, or social networks.

Second, higher prior uncertainty is associated with higher belief stickiness. The noisier the history of signals the agents received in the past, the more accurate the new signal will be *relative* to this stock of existing information. As a result, the agent assign more weight to the new signal in forming posterior beliefs, i.e. belief stickiness is lower.

We make two related observations. First, these implication holds similarly in models with endogenous information or rational inattention, where the signal noise $\sigma_{e,t}^2$ becomes a choice variable (Sims, 2003, 2006; Mackowiak and Wiederholt, 2009; Maćkowiak et al., 2023). Equation (8) shows that signal noise determines belief stickiness, regardless of whether it is endogenously chosen or externally determined. Second, while derived under the assumption of rational expectations, the comparative statics of equation (8) remain valid in many frameworks that build on the Bayesian benchmark but allow for behavioral deviations—such as diagnostic expectations (Bordalo et al., 2018, 2020; Bianchi et al., 2024a) and overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025). In contrast, these results do not hold in models where the updating gain G_t is constant and independent of uncertainty, such as sticky information (Mankiw and Reis, 2002), adaptive learning with a constant gain (Orphanides and Williams, 2007; Eusepi and Preston, 2011), and the benchmark version of behavioral inattention (Gabaix, 2019).

3.1.2 Belief stickiness and sources of uncertainty

We empirically test the two implications of the Bayesian belief updating framework in equation (8). First, higher prior uncertainty implies lower belief stickiness. Second, higher new information noise implies higher belief stickiness. To do this, we need to construct a measure of prior uncertainty and new information noise.

Prior uncertainty We construct a proxy for prior belief uncertainty from the belief updating model in equation (3) and (4), which implies

$$\sigma_{i,j,t+h,t-1}^2 = \rho^2 \sigma_{i,j,t+h-1,t-1}^2 + \sigma_u^2 \quad (9)$$

where $\sigma_{i,j,t+h-1,t-1}$ is the uncertainty measured from density forecasts provided by the same individual in the previous month, and $\rho = 0.946$ and $\sigma_u = 0.38$ are measured from inflation data and reported in Table A.1 In Appendix P, we replicate the analysis of this section using different parameters to construct priors and show that the results do not significantly depend on the values of ρ and σ_u .

Our measure of prior uncertainty is valid only in the absence of structural changes in the volatility of the underlying inflation process σ_u . An increase in (real or perceived) fundamental volatility, $\sigma'_u > \sigma_u$, would raise households' prior uncertainty in month t for a given level of posterior uncertainty observed in the previous month $t-1$. The severe economic

Table 1: Belief stickiness and uncertainty

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$	(4) $1 - G_{j,t}$
$\ln(\sigma_{j,t+h,t-1})$	-0.199*** (0.034)		-0.410*** (0.029)	-0.264*** (0.047)
$\ln(\hat{\sigma}_{e,j,t})$		0.301*** (0.021)	0.359*** (0.027)	0.354*** (0.029)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.12	0.47	0.63	0.65
Observations	768	768	768	768

Notes: This table reports the estimated coefficient from regression (11). Standard errors (in parentheses) are computed with a two-level bootstrap, within and across (j,t) groups. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

and social disruptions during the COVID period may have triggered such a structural change. For this reason, we restrict the analysis in this section to the pre-COVID period. In Section 3.2, we return to this issue and use our framework to provide evidence for such a structural break in fundamental volatility during COVID.

New information noise A main limitation of using natural variation in survey expectations compared to an experimental setting is that the econometrician does not directly observe the households' new information noise. To address this, we leverage our general framework to estimate it from the survey data.

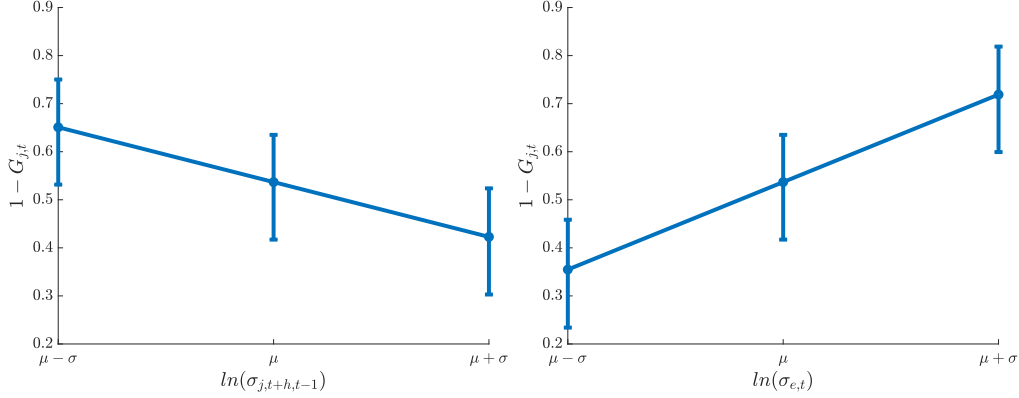
We use our estimate of belief stickiness, together with our measures of prior and posterior uncertainty, to compute the new information noise from (4).²¹

$$\hat{\sigma}_{e,j,t}^2 = \frac{\sigma_{j,t+h,t}^2 - \hat{\beta}_{j,t}^2 \sigma_{j,t+h,t-1}^2}{(1 - \hat{\beta}_{j,t})^2}, \quad (10)$$

where $\sigma_{j,t+h,t} \equiv \int^i \sigma_{i,j,t+h,t} di$ is the mean of individual posterior uncertainty in group j , and similarly for $\sigma_{j,t+h,t-1} \equiv \int^i \sigma_{i,j,t+h,t-1} di$. We drop the observations $\hat{\sigma}_{e,j,t}$ lower than zero, around 5% of the sample, and trim the top percentile to avoid outliers from measurement errors.

²¹We notice that equation (4) is part of our general framework and it does not impose any model of belief stickiness, meaning it does not make any assumption on G_t^j .

Figure 4: Belief stickiness and uncertainty



Notes: The figure represents graphically the estimated coefficients from column (4) of Table 1. It shows the relationship between belief stickiness and prior uncertainty (left panel) and new information noise (right panel).

Uncertainty and belief stickiness Next, we examine whether the relationship between our estimates of belief stickiness and measures of prior uncertainty and new information noise aligns with the Bayesian RE framework (8).

We regress belief stickiness on this new measure of new information noise. That is, we run the regression

$$1 - G_{j,t} = \alpha + \delta_1 \ln(\Sigma_{j,t+h,t-1}) + \delta_2 \ln(\hat{\sigma}_{e,j,t}) + \gamma_t + \gamma_j + \epsilon_{j,t}, \quad (11)$$

where $1 - G_{j,t} = \hat{\beta}_{j,t}$ is the group-month belief stickiness estimated in regression (6). We include time and group-level fixed effects. To account for the errors related to the generated regressor $\hat{\sigma}_{e,j,t}$, we bootstrap the standard errors with 10000 draws both within time-groups (regression 6) and across time-groups (regression 11).

We test two hypotheses. First, for a given new information noise, higher prior uncertainty leads to lower belief stickiness: $\delta_1 < 0$. Second, for a given prior uncertainty, higher new information noise leads to lower belief stickiness: $\delta_2 > 0$.

Our results, reported in Table 1, strongly support both predictions. Column (4) shows that new information noise is positively associated with belief stickiness, while the effect of prior uncertainty is negative. Figure 4 illustrates these effects: a one-standard-deviation increase in prior uncertainty reduces belief stickiness by about 0.1 ($\approx 20\%$), while a similar increase in new information noise raises it by about 0.15 ($\approx 30\%$). In short, belief stickiness is robustly negatively related to prior uncertainty and positively related to new information noise, consistent with the Bayesian framework.

A natural concern is that regression (11) may suffer from mechanical correlation, since we use our estimates of belief stickiness $\hat{\beta}_{j,t}$ to both construct $\hat{\sigma}_{e,j,t}$ and as the dependent variable. We note that equation (10) does not imply that $\hat{\sigma}_{e,j,t}$ depends on belief stickiness, but rather that we use our estimates to remove belief stickiness from posterior uncertainty and properly isolate $\sigma_{e,j,t}$. Still, measurement error might still introduce a bias. Three observations mitigate this concern. First, we removed the top percentile of $\hat{\sigma}_{e,j,t}^2$ to limit the influence of outliers. Second, measurement error does not affect our proxy for prior uncertainty $\sigma_{i,j,t+h,t-1}$, which still shows the expected negative effect in Column (1). Third, in Appendix G, we replicate the analysis using two alternative proxies for new information noise—posterior uncertainty and realized forecast errors (controlling for prior uncertainty)—and obtain qualitatively similar results. While these alternatives cannot perfectly isolate new information noise, they reinforce the robustness of our main findings.²²

In conclusion, we find that consumers are qualitatively Bayesian: they place more weight on new information when their priors are less certain, but exhibit more rigidity when the new information itself is uncertain. This pattern is a necessary condition for rational expectations, but not a sufficient one, since several behavioral models share the same qualitative predictions. In the next step, we therefore examine whether households are also quantitatively Bayesian, by testing whether the magnitude of their updating is consistent with the rational expectations benchmark.

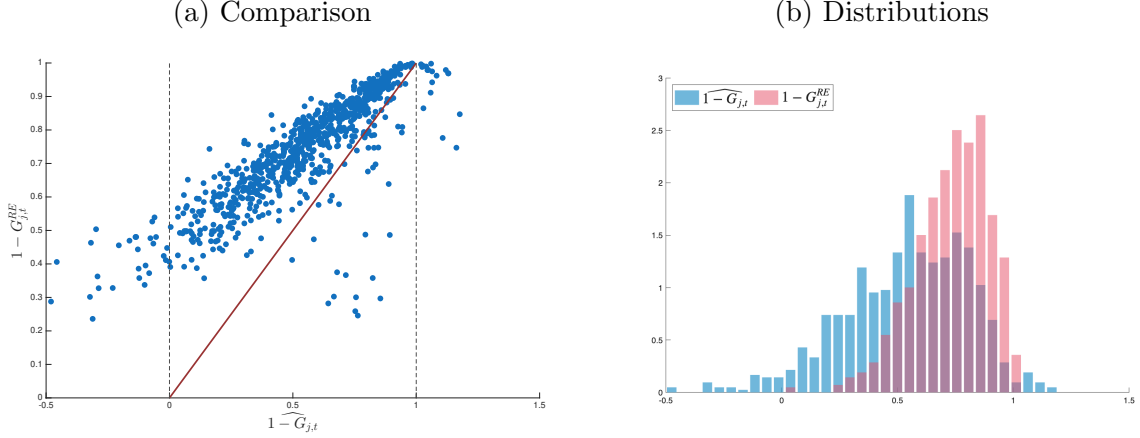
3.1.3 Are households rational?

We use our proxies for prior uncertainty and new information noise to construct the counterfactual belief stickiness in the Rational Expectation framework. We notice that, while we used our empirical estimates of belief stickiness to recover a measure for new information noise, we have not made any assumptions on what determines belief stickiness. This allows us to compare the belief stickiness measured in the data with the one a consumer would display if he updated rationally given the observed prior uncertainty and new information noise. From equation (8),

$$1 - \tilde{G}_{j,t}^{RE} = \frac{\hat{\sigma}_{e,t}^{j,2}}{\hat{\sigma}_{e,t}^{j,2} + \sigma_{j,t+h,t-1}^2}. \quad (12)$$

²²Recent studies suggest that households’ belief updating may depend on the level of inflation (Pfäuti, 2023; Weber et al., 2025). To assess this possibility, Table A.8 reports our estimates when controlling for annual, monthly, and lagged monthly inflation. We find no significant effect of inflation on belief stickiness in the pre-COVID sample.

Figure 5: Counterfactual RE and estimated belief stickiness



Notes: The left panel plots on the x-axis the estimated belief stickiness $\hat{\beta}_{j,t} = 1 - \widehat{G}_{j,t}$ from regression (6) and on the y-axis the implied rational expectation belief stickiness $1 - G_{j,t}^{RE}$ from equation (8). If the observation lies above the 45-degree red line, then the implied RE belief stickiness is larger than the estimated one. The right panel plots the two distributions, showing that the distribution of the implied RE belief rigidities shifted to the right of the estimated ones. The difference between the two means is ≈ 0.18 , with standard error ≈ 0.007 (two-level bootstrap, within and across (j,t) groups). Sample Jun13-Feb20.

Figure 5 reports the results for each group-month in the pre-COVID sample.²³

We find that households consistently display lower belief stickiness than predicted by the RE benchmark. In Figure 5a, most estimates lie above the 45-degree line, indicating that households assign more weight on new information than the rational counterfactual. Similarly, Figure 5b shows that the distribution of estimated stickiness centers around 0.6, compared with about 0.75 under RE. In other words, households in the SCE assign roughly 25% more weight to new information than the rational benchmark would imply, systematically overreacting relative to the Rational Expectations framework.²⁴

Estimating a belief formation model Next, we investigate potential biases in consumer belief formation by testing the following functional form for estimated belief stickiness:

$$1 - G_{j,t} = 1 - \left(\bar{g} + (1 + \theta) \frac{\sigma_{j,t+h,t-1}^2}{(1 - \alpha) \hat{\sigma}_{e,t}^{j,2} + \sigma_{j,t+h,t-1}^2} \right) \quad (13)$$

²³Because we extract new information noise from individual subjective uncertainty, our measure reflects *perceived* information noise, which may differ from the *actual* noise. Thus, a belief stickiness equal to (12) is a necessary but not sufficient condition for rationality. Full rationality would also require that perceived and actual information accuracy coincide. We show that even this necessary condition is rejected in the data.

²⁴Figure A.3 replicates the comparison excluding respondents in the lowest tercile of survey tenure, i.e., considering only consumers participating in the survey for longer periods. The results are virtually unchanged.

Table 2: GMM estimation

	(1)	(2)	(3)	(4)
θ	0.642*** (0.034)			0.306** (0.128)
α		0.619*** (0.010)		0.289* (0.162)
$\bar{g}$			0.185*** (0.007)	0.027 (0.018)
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Observations	768	768	768	768

Notes: This table reports the estimated coefficients θ, α and $\bar{g}$ from objective function (13) with GMM, weighting by (j, t) group size. $\alpha \neq 0$ and $\theta \neq 0$ capture different forms of overreaction documented in the literature, and $\bar{g}$ allows for a constant component of belief stickiness. Standard errors (in parentheses) are computed with a two-level bootstrap, within and across (j, t) groups. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

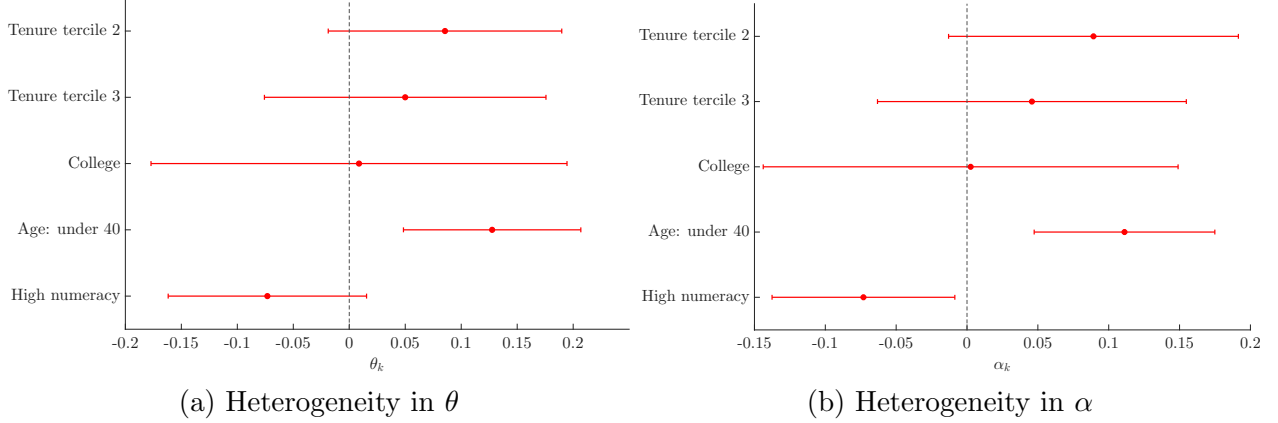
We introduce three parameters that can generate a wedge between the estimated belief stickiness and the RE benchmark in (12): $\bar{g}$, θ , and α , each capturing different behavioral departures. If all are zero, the model collapses to the Rational Expectations Kalman gain. First, $\alpha \neq 0$ captures a relative mis-weighting of new information noise versus prior uncertainty compared to G^{RE} . When $0 < \alpha < 1$, this bias resembles overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025).²⁵ Second, $\theta \neq 0$ (with $\alpha = 0$) represents a proportional distortion of the Kalman gain, akin to diagnostic expectations (Bordalo et al., 2018, 2020).²⁶ Third, $\bar{g}$ allows for a constant component of stickiness unrelated to information accuracy. We estimate the three parameters by generalized method of moments, minimizing the distance between (13) and our empirical stickiness estimates. Table 2 reports the results.

We find $\alpha > 0$ and $\theta > 0$, while $\bar{g}$ is not significantly different from zero. Both results imply that households overweight new relative to prior information. Thus, while households are *qualitatively* Bayesian — updating more when prior beliefs are uncertain and less when signals are noisy — they are not *quantitatively* Bayesian. Instead, they consistently place too much weight on new information relative to the rational benchmark, leading to systematic

²⁵An important difference between the bias we test here and overconfidence is that in the latter framework agents overweight new information because they underestimate its noise. Instead, we extract the new information noise from consumers’ subjective beliefs. Therefore, we estimate how much they overweight their subjective new information noise.

²⁶An important difference between the bias we test here and diagnostic expectation is that in the latter framework the bias only applies to the first moment of belief and not to the second moment. In our framework, the same belief stickiness is applied to first and second moments of beliefs, equation (3) and (4) (Bordalo et al., 2020). Bianchi et al. (2024b) proposes a “smooth” DE setting where the bias also affects the second moment of beliefs.

Figure 6: Heterogeneity in overreaction bias



Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 13. Table A.7 reports the detailed results. The right panel plots the estimated interactions with socioeconomic dummies and α in 13. Table A.6 reports the detailed results. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across (j, t) groups. Sample Jun13-Feb20.

overreaction.

Heterogeneity in overreaction bias In Section 2.2.1, we document heterogeneity in belief adjustment, with high numeracy-skilled and older households exhibiting higher belief stickiness. This result might be due either to an optimal choice of placing less weight on noisier new information, or to a lower overreaction bias. We now investigate this heterogeneity in overreaction biases.

We interact the overreaction parameters in eq (13) with dummies for socioeconomic characteristics and estimate $\theta = \theta_0 + \sum_{k=1}^5 \theta_k D_k$, where $D_k = 1$ is the group belong to the socioeconomic characteristics k and θ_k measure the impact of this characteristic on the overreaction parameter θ . Figure 6a reports the estimated interactions, while the detailed results are in Table A.7. We find that younger consumers display a higher overreaction bias, while consumers with high numeracy skills display a lower overreaction bias. We find similar results when interacting socioeconomic dummies with α in Figure 6b. These results suggest that the higher belief stickiness of older and high-numeracy households does not stem from lower information quality, but from a lower overreaction bias. Importantly, the intuitive link between overreaction and socioeconomic traits also reduces concerns that our findings are driven by measurement error.

Our approach provides a novel test of rationality in survey expectations. By relating belief updating to subjective uncertainty, we can detect systematic biases and their heterogeneity

across households, even in surveys with limited time dimensions such as the SCE. The results underscore the importance of distinguishing whether belief stickiness reflects information quality or behavioral distortions, with direct implications for household responses to shocks and for the effectiveness of policy communication.

3.2 Unpacking the sources of uncertainty post-COVID

In this section, we combine our estimates of belief stickiness with the belief formation model estimated in the pre-pandemic economy to decompose the sources of the uncertainty surge in the post-pandemic economy.

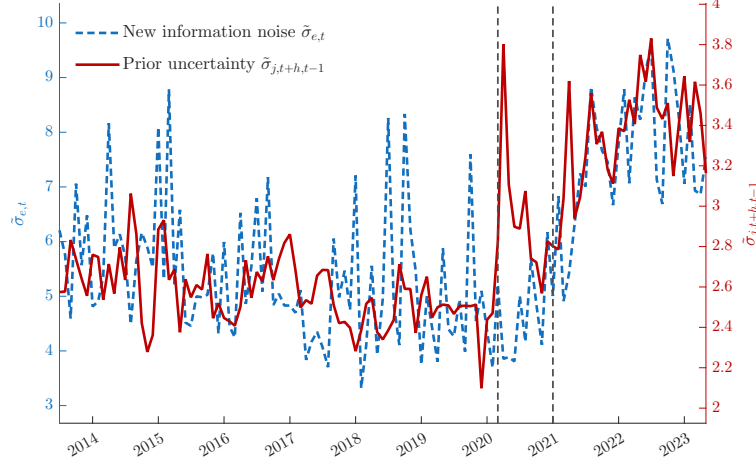
In the pre-COVID sample, we proxy prior uncertainty using lagged posterior uncertainty adjusted for the estimated fundamental volatility. When structural shifts occur, however, this proxy is no longer valid, as volatility may differ from historical estimates. To address this, we exploit the belief formation model in (13), estimated on pre-COVID data, together with observed belief stickiness and posterior uncertainty, to recover the underlying sources of uncertainty after 2020. This approach not only decomposes the surge in uncertainty but also provides evidence for structural breaks in the fundamental volatility.

Our general framework (4) implies that posterior uncertainty depends positively on both new information noise $\sigma_{e,t}$ and prior uncertainty $\sigma_{t+h,t-1}$. In turn, our estimated belief updating model (13) shows that belief stickiness increases with the former but decreases with the latter. Solving this system of two equations yields

$$\begin{aligned}\sigma_{e,j,t}^2 &= \frac{(\theta + 1 - G_{j,t})\sigma_{j,t+h,t}^2}{(1 - G_{j,t})^2(1 - \alpha)G_{j,t} + G_{j,t}^2(\theta + 1 - G_{j,t})} \\ \sigma_{j,t+h,t-1}^2 &= \frac{(1 - \alpha)G_{j,t}}{(\theta + 1 - G_{j,t})}\sigma_{e,j,t}^2.\end{aligned}\tag{14}$$

We can thus recover new information noise $\tilde{\sigma}_{e,j,t}^2$ and prior uncertainty $\tilde{\sigma}_{j,t+h,t-1}$ as functions of the estimated belief stickiness $\widehat{1 - G_{j,t}}$, posterior uncertainty $\sigma_{j,t+h,t}$, and the overreaction parameters $\hat{\alpha}, \hat{\theta}$. Figure 7 reports the time series of the group-averaged estimates and reveals distinct patterns for prior uncertainty and new information noise across the pandemic and high-inflation periods. Figure A.4 presents the same decomposition under the Rational Expectations benchmark ($\alpha = 0, \theta = 0$), with similar results.

Figure 7: Decomposing uncertainty: prior and new information noise



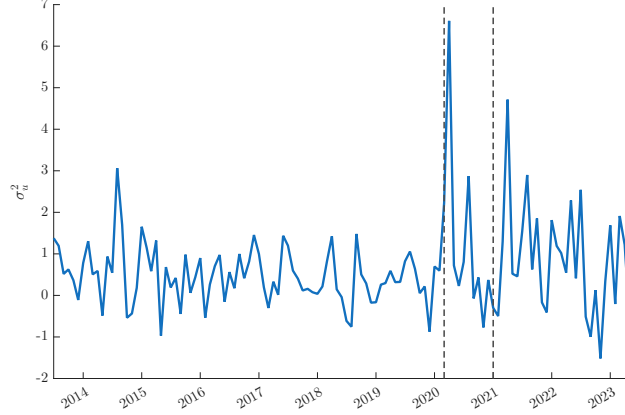
Notes: The figure plots the estimates of new information noise (left y-axis) and prior uncertainty (right y-axis) from equation (14), with the estimates of estimates of belief stickiness $1 - \widehat{G}_{j,t}$, posterior uncertainty $\sigma_{j,t+h,t}$ and $\hat{\alpha}, \hat{\theta}$.

Post-COVID From 2021, after the initial months of COVID-19, we document a sharp rise in new information noise (Figure 7). The increase in noise *relative* to prior uncertainty accounts for the positive correlation between belief uncertainty and stickiness observed in Figure 3. Later, although new information noise remains elevated, its relative weight compared to prior uncertainty returns to pre-COVID levels. This shift helps explain why belief stickiness resembles pre-COVID values, even as overall belief uncertainty is higher in the post-COVID period.

Higher information noise may result from consumers facing greater costs in gathering information or a reduced supply of information from newspapers, television, and social networks. This period, beginning in February 2021, is also marked by rising inflation and increased media coverage. At first glance, this may seem inconsistent with the observed increase in noise in new information about inflation. However, greater media coverage does not necessarily imply more accurate information dissemination. Instead, it may reflect efforts to manage heightened uncertainty driven by the increasing difficulty of predicting inflation, in a moment when even economists disagreed on how long the high-inflation period would have lasted (Markovitz, 2021).

COVID outbreak Interestingly, we find a sharp increase in prior uncertainty at the COVID-19 outbreak, in March 2020. This follows from the contemporaneous spike in belief uncertainty and decline in belief stickiness displayed in Figure 3. This might reveal a struc-

Figure 8: Model-implied fundamental volatility



Notes: The figure plots the estimates of fundamental volatility $\sigma_{u,t}^2 = \tilde{\sigma}_{t+h,t-1}^2 - \rho^2 \sigma_{t+h-1,t-1}^2$.

tural change in the underlying volatility of the economy in the post-COVID era, which can be traced to two possible sources.

First, consider an increase in fundamental volatility $\sigma_u'^2 > \sigma_u^2$. As shown in equation (9), such higher volatility implies that prior information becomes obsolete, and therefore more uncertain, when forecasting the future, as the stochastic process of the fundamental becomes more unpredictable. We do not take a stand on what could have driven such an increase in fundamental volatility, as some determinants might have been specific to the COVID-19 case: the lethality of the virus, the capacity of healthcare systems to meet an extraordinary challenge, its economic consequences, the waiting time to develop a safe vaccine, et cetera (see Baker et al. (2020)).

We recover the level of fundamental volatility implied by our measure of prior uncertainty by inverting equation (9). Specifically, we compute $\sigma_{u,t}^2 = \tilde{\sigma}_{t+h,t-1}^2 - \rho^2 \sigma_{t+h-1,t-1}^2$, where $\tilde{\sigma}_{t+h,t-1} \equiv \int^j \tilde{\sigma}_{j,t+h,t-1} dj$ and $\sigma_{j,t+h-1,t-1} \equiv \int^j \sigma_{j,t+h-1,t-1} dj$. Figure 8 shows that fundamental volatility is broadly stable in the pre-COVID period but rises sharply at the onset of the pandemic, driving the observed increase in prior uncertainty.

Second, consider a loss of trust in the central bank's ability to maintain its inflation target. In this stylized setting, one can think of the belief about the long-run mean of inflation becoming uncertain, $\mu_x \sim N(\bar{\pi}, \sigma_\mu^2)$. Prior uncertainty then becomes

$$\sigma_{j,t+h,t-1}^2 = (1 - \rho)^2 \sigma_\mu^2 + \rho^2 \sigma_{j,t+h-1,t-1}^2 + \sigma_u^2. \quad (15)$$

A larger uncertainty about long-run mean inflation σ_μ^2 would increase prior uncertainty and, as a result, lower belief stickiness. This could be possibly due to lower trust in the central

bank’s ability to achieve its price stability objective. [Aikman et al. \(2024\)](#) measures the trust in the Federal Reserve from Twitter data and finds the lowest point at the COVID outbreak. Alternatively, it could be more generally due to uncertainty about economic policy implementation.²⁷

A regime shift in the autoregressive parameter ρ could also have played a role. However, detecting a structural change within such a short sample is challenging. In [Appendix I](#), we present rolling-window AR(1) regressions of monthly inflation and show that there is no strong evidence for a shift in the autoregressive parameter ρ around the COVID-19 period compared to previous periods.

Another possible force driving the change in belief stickiness during COVID-19 is the lockdown policy restrictions on movements implemented by policymakers to stop the spread of the virus. These restrictions might have lowered the cost of browsing for news and therefore contributed to this decrease in belief stickiness. We investigate this in the next section.

4 Lockdowns and belief formation

4.1 Impact of lockdowns on belief stickiness

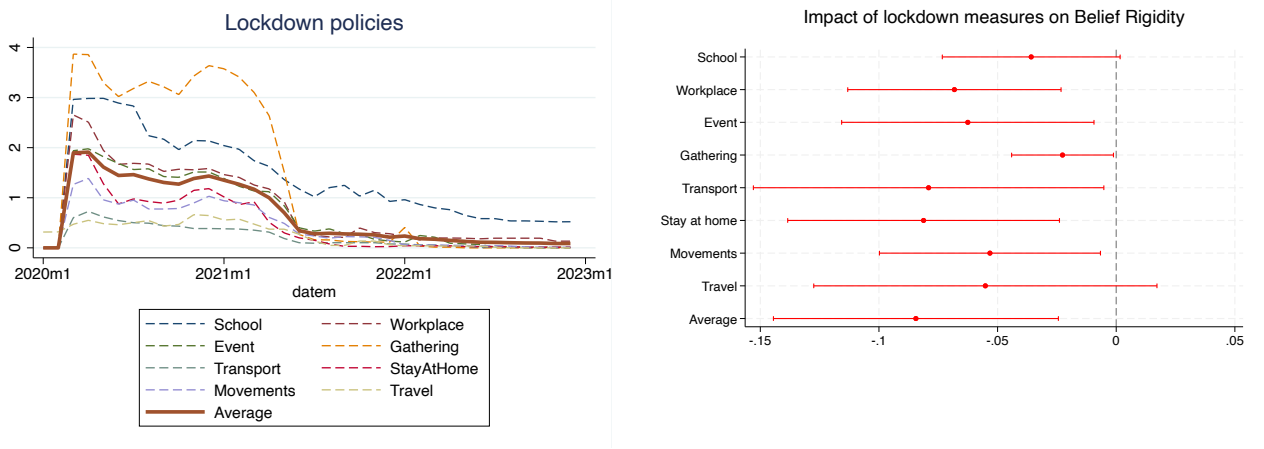
In this section, we investigate the role played by lockdown policies in driving the decline in belief stickiness we documented during the pandemic. The restrictions on movement and activity implemented to stop the spread of the virus might have played a role in shaping both households’ attention dynamic and their subjective uncertainty.

To analyze the impact of lockdown policies on belief stickiness, we use state-level lockdown stringency measures from the Oxford COVID-19 Government Response Tracker (OxCGRT) ([Hallas et al., 2021](#)). This tracker compiles data on the severity of closure and containment policies, such as school and workplace closings, restrictions on public events, gathering sizes, public transportation, stay-at-home mandates, and internal movement. These indicators are averaged into a summary lockdown measure, weighted by the share of vaccinated and non-vaccinated individuals in each state.²⁸ [Figure 9](#) reports the time series of the country-level average of each indicator. We also track local COVID-19 impact using per capita state-level

²⁷Figure [A.5](#) plots the Economic Uncertainty Index, constructed by [Baker et al. \(2016\)](#) and offers additional evidence for this implication. We plot the “news coverage component”, which measures the share of articles in US online newspapers that mention economic policy uncertainty. This index spikes up during the Covid outbreaks and decreases to the pre-Covid level in the following period, corroborating our results that the increase in belief uncertainty observed during Covid might be due to this policy uncertainty.

²⁸We provide more details on the index construction in the [Appendix K](#).

Figure 9: Belief stickiness and uncertainty



Notes: The left figure represents the average state-level lockdown policy intensity for different social activities, weighted by state population. The data source for lockdowns is the Oxford Covid-19 Government Response Tracker (OxCGRT). The right plot shows the impact of lockdown measures on our estimate of belief stickiness, β_2 in (16). Sample period: 2020M3-2022M12.

COVID-19 cases and deaths.

To estimate the effect of lockdown measures on belief stickiness, we interact prior inflation forecasts with the lockdown index and COVID-19 case and death measures. This approach allows us to isolate the impact of lockdown policies from the effects of COVID-19's direct health impact. We run the following regression:

$$\begin{aligned}
 E_t^i[x_{t+h}] = & \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \beta_2 E_{t-1}^i[x_{t+h}] \times Lockdown_{k,t} + \beta_3 Lockdown_{k,t} + \\
 & + E_{t-1}^i[x_{t+h}] \times CovidSeverity_{k,t} \Pi + CovidSeverity_{k,t} \Gamma + \\
 & + \lambda X_{i,t} + [CovidSeverity_{k,t} \quad Lockdown_{k,t}] \times \gamma_t + err_t^i,
 \end{aligned} \tag{16}$$

where $Lockdown_{k,t}$ contains the lockdown indexes, while $CovidSeverity_{k,t}$ contains the COVID cases and death in state k at date t . The key coefficient of interest, β_2 , captures how lockdown policies influence belief stickiness. We run the regression in the post-pandemic sample, from March 2020. Figure 9 reports the estimates of β_2 , while Table A.5 reports the detailed result.

We find that all lockdown indicators have a robust negative effect on belief stickiness. This result suggests that lockdown policies might have lowered the cost of collecting information for consumers, leading them to adjust their beliefs more than before. The results are robust to controlling for economic policy uncertainty (i.e., the newspaper-based Economic Policy Uncertainty, EPU), as reported in table A.10.

4.2 Impact of lockdowns on uncertainty

The documented negative impact of lockdown restrictions on belief stickiness is consistent with a decline in the cost of collecting information. From the lens of our general framework in Section 1, a lower marginal cost of information collection corresponds to a decrease in new information noise $\sigma_{e,t}^2$ (Maćkowiak et al., 2023; Pomatto et al., 2023). Higher new information accuracy would then not only lead to a decrease in belief stickiness, but also a decline in posterior uncertainty.²⁹ To test this empirically, we estimate the following regression at the US state level:

$$\sigma_{k,t+h,t} = \alpha + \beta Lockdown_{k,t} + CovidImpact'_{k,t} \Gamma + \delta \ln(EPU)_{k,t} + \gamma_j + err_{k,t} \quad (17)$$

where $\sigma_{k,t+h,t} = \int_{i \in k} \sigma_{i,t+h,t} di$ denotes the average posterior uncertainty for consumers in state k at time t . The key variable of interest is $Lockdown_{k,t}$, the average index of lockdown intensity. Table A.14 reports the estimated coefficients.

Results show a robust and negative effect of lockdown policies on posterior belief uncertainty, consistent with the idea that lower information costs allow for better information gathering. Economic policy uncertainty ($EPU_{k,t}$) increases posterior uncertainty, as expected.

These findings show that lockdown policies significantly reduced belief stickiness while also lowering posterior uncertainty, indicating that consumers perceived the information they gathered as more precise. This result is consistent with theoretical predictions about the effects of lower information costs on belief formation.³⁰

At the aggregate level, however, reduced information costs can account for the decline in belief stickiness during the pandemic, but cannot explain the simultaneous rise in uncertainty. As discussed in Section 3.2, only an increase in fundamental volatility can reconcile both empirical patterns.

²⁹An alternative possibility is that lower information costs led to higher, instead of lower belief uncertainty. This could be the case, for example, if consumers could learn about signals' accuracy only by acquiring more signals. In this case, a lower information cost would allow consumers to acquire more signals and learn about the increase in the signal's noise, which could explain both the lowering belief stickiness and the higher belief uncertainty.

³⁰Our findings are also consistent with a negative effect of lockdown policies on state-level fundamental uncertainty, which households may have extrapolated to aggregate conditions. Such a mechanism would likewise reduce perceived uncertainty and increase belief updating.

Table 3: Belief stickiness and lockdown measures

	(1)	(2)	(3)
	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$
<i>Lockdown</i>	-0.379*** (0.084)	-0.362*** (0.084)	-0.234*** (0.063)
$\sigma_{k,t+h-1,t-1}$			0.464*** (0.034)
$\ln(DeathsCOVID)$			0.017 (0.036)
$\ln(CasesCOVID)$			-0.028 (0.037)
$\ln(EPUNational)$			0.068** (0.032)
Constant	3.569*** (0.103)	3.557*** (0.052)	1.621*** (0.313)
State FEs	N	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0.05	0.28	0.43
Observations	1717	1717	1694

Notes: The table reports the results from regression 17. *EPUComposite* is the state-level economic policy uncertainty indicator from Baker et al. (2022). We control for state FEs. Covid controls are logged *DeathsCOVID* and *CasesCOVID*, which are the state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at State and time levels. *, **, and *** are 10%, 5%, and 1% confidence levels.

5 Conclusion

Uncertainty does not have a uniform effect on expectations; its consequences depend on its source. When uncertainty reflects less confidence in prior beliefs, households place greater weight on incoming information and update more. When uncertainty reflects noisier signals, updating diminishes and prior beliefs exert greater influence. This framework organizes the post-2020 evidence: the early pandemic is consistent with a shift in fundamentals that raised prior uncertainty and accelerated updating, whereas the subsequent high-inflation phase features rising uncertainty alongside greater stickiness, indicative of noisier information. The estimates accord with Bayesian comparative statics while revealing systematic overreaction in levels relative to a rational benchmark.

Our results have important policy-relevant implications. Because belief stickiness depends on the source of uncertainty, it can be used to identify whether it is driven by noisier signals—where clearer communication and information provision can help—or elevated fundamental volatility, where communication has limited traction and broader stabilization tools are required. With high-frequency expectations data, tracking stickiness offers a practical, real-time read on the uncertainty source to inform policy.

Our work also suggests several directions for future research. While this paper focuses on household expectations, understanding the determinants of information frictions on the production side of the economy is equally important to improve macroeconomic models, allowing for a more precise study of macroprudential policy and its dependence on different sources of uncertainty. In addition, while this paper studies the total adjustment of beliefs, future work can use this methodology to distinguish between intensive and extensive margins of belief adjustment. Finally, belief stickiness could be used empirically to examine the macroeconomic effects of different types of uncertainty.

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Online Appendix

A Belief formation models

The framework in equation (3) encompasses not only Rational Expectations but also alternative models of belief formation, particularly those that incorporate frictions in processing new information. Some examples are:

- Rational expectations: $G_t^{RE} = \frac{\tau_t}{\tau_t + \Sigma_{t+h,t-1}^{-1}}$, where $\Sigma_{t+h,t-1} \equiv \text{var}(x_{t+h} - E_{t-1}^i[x_{t+h}])$ is the prior variance (Sims, 2003; Woodford, 2001; Mackowiak and Wiederholt, 2009). In the case of full-information, the signal is perfectly informative, $\tau_t \rightarrow \infty$, and therefore $G_t = 1$.
- Diagnostic expectation (Bordalo et al., 2018, 2020): households overreact to new information according to $\theta > 0$, therefore $G_t = (1 + \theta)G_t^{RE}$.
- Overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025): households perceived signal accuracy as more accurate, $\tilde{\tau}_t > \tau_t$, and therefore $G_t = \frac{\tilde{\tau}_t}{\tilde{\tau}_t + \Sigma_{t+h,t-1}^{-1}} > G_t^{RE}$.
- Sticky information (Mankiw and Reis, 2002): household has a probability $1 - \lambda$ of fully updating her beliefs $G_t = 1$, and λ of not updating their belief at all, $G_t = 0$.
- Memory frictions (Azeredo da Silveira et al., 2024): agents pay a cost not only to process new information but also past information, meaning that prior accuracy decay with time. Then, G_t would internalize both the noise of new information and the memory friction.
- Learning with constant gain (Eusepi and Preston, 2011): households learn about the model's parameters in each period using a constant gain, so that they never learn completely.
- Behavioral inattention (Gabaix, 2019): agents pay limited attention to some variable x for non-rational reasons. It can map in our framework if $G_t = m_t$ and the "default" belief is the prior.

By contrast, the structure in (3) is not consistent with belief models in which agents lack knowledge of the true economic structure and instead rely on alternative perceived laws of motion or simple heuristics, such as natural expectations (Fuster et al., 2010), over- or under-extrapolation (Angeletos et al., 2021), or narrative-based beliefs (Andre et al., 2024).

B Point estimates and subjective distribution of inflation in the SCE

Q9c

And in your view, what would you say is the percent chance that, **over the 12-month period between August 2015 and August 2016 ...**

Instruction H4.

the rate of inflation will be 12% or higher	___ percent chance
the rate of inflation will be between 8% and 12%	___ percent chance
the rate of inflation will be between 4% and 8%	___ percent chance
the rate of inflation will be between 2% and 4%	___ percent chance
the rate of inflation will be between 0% and 2%	___ percent chance
the rate of deflation (opposite of inflation) will be between 0% and 2%	___ percent chance
the rate of deflation (opposite of inflation) will be between 2% and 4%	___ percent chance
the rate of deflation (opposite of inflation) will be between 4% and 8%	___ percent chance
the rate of deflation (opposite of inflation) will be between 8% and 12%	___ percent chance
the rate of deflation (opposite of inflation) will be 12% or higher	___ percent chance

Total

100

C Alternative measure of belief uncertainty

A drawback of using bins questions to measure the individual density forecast is that the intervals considered in the bins might too wide or too narrow to capture the whole belief distribution fully. The bins of the Survey of Consumer Expectations range from -12% to 12% in steps of 2 to 4 percentage points. During high inflation periods, such as the post-Covid months, consumers might attribute a large probability on inflation realization above this upper bound, which could lead to inaccuracies in measuring their true belief distribution from bins question.

To address this concern, we compare our benchmark measure with an alternative measure of belief uncertainty based on the rounding of point forecasts, as in [Binder \(2017\)](#). In each month we compute the share of respondents that provide a forecast multiple of 5. This uncertainty measure is based on the Round Numbers Suggest Round Interpretation (RNRI) principle, which suggests that round numbers are frequently used to convey that a quantitative expression should be interpreted as imprecise ([Krifka, 2007](#)).

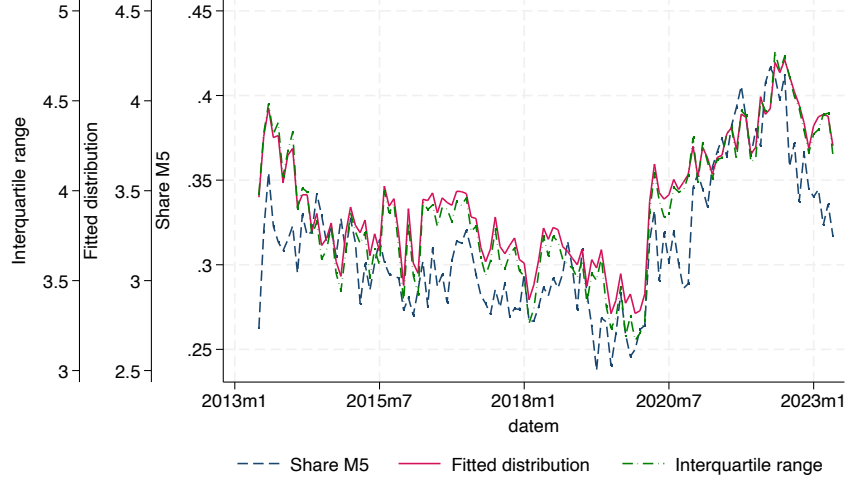


Figure A.1: Uncertainty measures

Notes: This plot shows different measures of subjective uncertainty over time. The blue dashed line is the share of respondents that provide a forecast multiple of 5 from [Binder \(2017\)](#), the red solid line is the average subjective standard deviation provided by the SCE, and the green dotted line is the average subjective interquartile range provided by the SCE. Sample period: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure [A.1](#) plots the measure based on rounding (‘Share M5’, from [Binder \(2017\)](#)) together with the average uncertainty measures from the bins survey question: the standard deviation of the fitted distribution and the interquartile range. The rounding uncertainty measure closely tracks the other two and exhibits the same pattern during and after the COVID period.

D Inflation autoregressive process

We estimate [\(1\)](#) with the following regression on the 12-month inflation rate x_t :

$$x_t = (1 - \rho)\mu_x + \rho x_{t-1} + u_t, \quad (\text{A.1})$$

and we recover the long-run mean $\mu_x = \frac{\hat{\alpha}}{(1-\hat{\rho})}$ and the standard deviation of the fundamental noise $\sigma_u = \text{std}(\hat{u}_t)$. We run the regression on different subsamples, and the resulting coefficients are quite stable.

Table A.1: Inflation autoregressive proces

	(1) <i>Inflation</i>	(2) <i>Inflation</i>	(3) <i>Inflation</i>	(4) <i>Inflation</i>
<i>L.Inflation</i>	0.946*** (0.016)	0.975*** (0.013)	0.968*** (0.007)	0.934*** (0.023)
μ_x	2.3	2.82	2.47	2.15
σ_u	.38	.4	.38	.44
Sample	Jan90-Feb20	Jan90-May23	Jan80-Feb20	Jan00-Feb20
Observations	361	394	481	241

Notes: * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

E Additional tables

Table A.2: Descriptive Statistics

	Mean	SD	Min	Max	N
Beliefs					
<i>Forecast</i>	5.24	8.68	-50	75	131007
<i>Revision</i>	-0.21	7.27	-100	110	95098
<i>Post Uncert</i>	2.94	3.06	0	22	133362
<i>Post Uncert IQR</i>	3.32	3.52	0	28	133362
Socioeconomic characteristics					
<i>College_{it}</i>	0.89	0.31	0	1	135669
<i>Income 50kto100k_{it}</i>	0.35	0.48	0	1	134293
<i>Income Over100k_{it}</i>	0.30	0.46	0	1	134293
<i>Income Under50k_{it}</i>	0.34	0.47	0	1	134293
<i>High Numeracy_{it}</i>	0.74	0.44	0	1	135610
<i>Female_i</i>	0.47	0.50	0	1	135606
<i>Age_{it}</i>	50.57	15.25	17	94	135549
<i>White_i</i>	0.85	0.35	0	1	135663
<i>Tenure_{it}</i>	5.62	3.39	1	16	135669

Notes: This table provides descriptive statistics for beliefs and household socioeconomic characteristics derived from the Survey of Consumer Expectations (SCE). The sample period is 2013M6-2023M5.

Table A.3: Heterogeneity in Belief Updating

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$	(5) $E_t^i[x_{t+h}]$	(6) $E_t^i[x_{t+h}]$	(7) $E_t^i[x_{t+h}]$	(8) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.573*** (0.011)	0.455*** (0.017)	0.526*** (0.023)	0.603*** (0.015)	0.561*** (0.019)	0.524*** (0.015)	0.557*** (0.024)	0.396*** (0.041)
<i>Tenure Tercile</i> =2 $\times E_{i,t-1}[x_{t+h}]$		0.120*** (0.021)						0.116*** (0.020)
<i>Tenure Tercile</i> =3 $\times E_{i,t-1}[x_{t+h}]$		0.234*** (0.020)						0.223*** (0.021)
<i>College</i> _{it} =1 $\times E_{i,t-1}[x_{t+h}]$			0.053** (0.025)					0.034 (0.025)
<i>Age Over</i> 60=1 $\times E_{i,t-1}[x_{t+h}]$				-0.023 (0.022)				-0.038* (0.023)
<i>Age Under</i> 40=1 $\times E_{i,t-1}[x_{t+h}]$				-0.085*** (0.025)				-0.079*** (0.024)
<i>Income Over</i> 100k=1 $\times E_{i,t-1}[x_{t+h}]$					-0.011 (0.033)			-0.020 (0.032)
<i>Income Under</i> 50k=1 $\times E_{i,t-1}[x_{t+h}]$					-0.003 (0.022)			0.015 (0.021)
<i>High Numeracy</i> =1 $\times E_{i,t-1}[x_{t+h}]$						0.075*** (0.018)		0.060*** (0.018)
<i>Female</i> =1 $\times E_{i,t-1}[x_{t+h}]$							-0.025 (0.022)	-0.009 (0.022)
<i>White</i> =1 $\times E_{i,t-1}[x_{t+h}]$							0.033 (0.022)	0.018 (0.022)
Constant	2.038*** (0.053)	1.975*** (0.051)	2.048*** (0.053)	2.038*** (0.053)	2.114*** (0.055)	2.042*** (0.050)	2.050*** (0.054)	2.068*** (0.051)
Year-Month FEs	Y	Y	Y	Y	Y	Y	Y	Y
Non-interacted variables	N	Y	Y	Y	Y	Y	Y	Y
Year-Month $\times$ variables	N	Y	Y	Y	Y	Y	Y	Y
Sample	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23
Adjusted R-squared	0.33	0.33	0.33	0.33	0.33	0.34	0.33	0.35
Observations	95098	95098	95098	95062	94164	95070	95059	94101

Notes: Standard errors (in parentheses) are clustered at the individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Table A.4: Descriptive Statistics

	Mean	SD	Min	Max	N
Lockdown policies					
<i>School</i>	1.45	0.96	0	3	35859
<i>Workplace</i>	0.82	0.91	0	3	35859
<i>Event</i>	0.72	0.79	0	2	35859
<i>Gathering</i>	1.44	1.78	0	4	35859
<i>Transport</i>	0.25	0.47	0	2	35859
<i>StayAtHome</i>	0.48	0.67	0	2	35859
<i>Movements</i>	0.45	0.66	0	2	35859
<i>Travel</i>	0.24	0.58	0	2	35859
<i>CasesCOVID</i>	0.01	0.01	0	0.103	35859
<i>DeathsCOVID</i>	0.00	0.00	0	0.00108	35859
Economic Policy Uncertainty					
<i>EPUState</i>	1.98	1.88	0	14.66	40756
<i>EPUNational</i>	1.97	1.53	0	15.63	40756
<i>EPUComposite</i>	3.23	2.47	0.151	19.64	40756

Notes: This table provides descriptive statistics for lockdown policy intensity (from [Hale et al. \(2020\)](#)) and economic policy uncertainty (from [Baker et al. \(2022\)](#)). The sample period is 2020M3-2023M5.

Table A.5: Belief stickiness and lockdown measures

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$	$E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.586*** (0.130)	0.616*** (0.113)	0.633*** (0.119)	0.610*** (0.116)	0.559*** (0.122)	0.626*** (0.112)	0.565*** (0.121)	0.548*** (0.124)	0.597*** (0.110)	0.655*** (0.114)
$E_{i,t-1}[x_{t+h}] \times \ln(DeathsCOVID)$	-0.030* (0.016)	-0.020 (0.016)	-0.019 (0.017)	-0.025 (0.017)	-0.033* (0.018)	-0.020 (0.015)	-0.032* (0.018)	-0.036* (0.019)	-0.020 (0.015)	-0.016 (0.016)
$E_{i,t-1}[x_{t+h}] \times \ln(CasesCOVID)$	0.046** (0.019)	0.033* (0.017)	0.038** (0.018)	0.046** (0.019)	0.052** (0.020)	0.038** (0.015)	0.051** (0.020)	0.057** (0.021)	0.032* (0.016)	0.034* (0.017)
$E_{i,t-1}[x_{t+h}] \times School$	-0.036* (0.019)								0.009 (0.025)	
$E_{i,t-1}[x_{t+h}] \times Workplace$		-0.068*** (0.023)							-0.069** (0.034)	
$E_{i,t-1}[x_{t+h}] \times Event$			-0.063** (0.027)						-0.001 (0.040)	
$E_{i,t-1}[x_{t+h}] \times Gathering$				-0.023** (0.011)					0.020 (0.015)	
$E_{i,t-1}[x_{t+h}] \times Transport$					-0.079** (0.038)				-0.032 (0.038)	
$E_{i,t-1}[x_{t+h}] \times StayAtHome$						-0.081*** (0.029)			-0.050 (0.042)	
$E_{i,t-1}[x_{t+h}] \times Movements$							-0.053** (0.024)		0.003 (0.038)	
$E_{i,t-1}[x_{t+h}] \times Travel$								-0.055 (0.037)	-0.018 (0.042)	
$E_{i,t-1}[x_{t+h}] \times Lockdown$										-0.084*** (0.031)
Constant	2.240*** (0.086)	2.252*** (0.079)	2.255*** (0.082)	2.250*** (0.084)	2.252*** (0.087)	2.248*** (0.080)	2.252*** (0.086)	2.244*** (0.089)	2.248*** (0.077)	2.256*** (0.081)
Year-Month FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year-Month $\times$ Variables	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0	0	0	0	0	0	0	0	0	0
Observations	24845	25626	25626	25626	25626	25626	25626	25626	25626	25626

Notes: Variables *School* to *Travel* measure lockdown policies intensity for different social activities, from the Oxford Covid-19 Government Response Tracker (OxCGRT). *Lockdown* is the average of the other lockdown indicators. We control for year-month fixed effects, and for socioeconomic characteristics, such as education, income, age, gender, race, and tenure. Covid Controls are $\ln(DeathsCOVID)$ and $\ln(CasesCOVID)$, which are the log state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Table A.6: Estimated overreaction α

	(1)
θ	0.317** (0.140)
α	0.244 (0.242)
$\alpha \times \textit{Tenuretercile2}$	0.089 (0.062)
$\alpha \times \textit{Tenuretercile3}$	0.046 (0.066)
$\alpha \times \textit{College}$	0.003 (0.089)
$\alpha \times \textit{Age : under40}$	0.111*** (0.039)
$\alpha \times \textit{Highnumeracy}$	-0.073* (0.039)
$\bar{g}$	0.030* (0.018)
Sample	Jun13–Feb20
Observations	768

*Notes:*Table A.7: Estimated overreaction θ

	(1)
α	0.287* (0.169)
θ	0.272** (0.136)
$\theta \times \textit{Tenuretercile2}$	0.086 (0.063)
$\theta \times \textit{Tenuretercile3}$	0.050 (0.076)
$\theta \times \textit{College}$	0.009 (0.113)
$\theta \times \textit{Age : under40}$	0.128*** (0.048)
$\theta \times \textit{Highnumeracy}$	-0.073 (0.054)
$\bar{g}$	0.028* (0.017)
Sample	Jun13–Feb20
Observations	768

Notes:

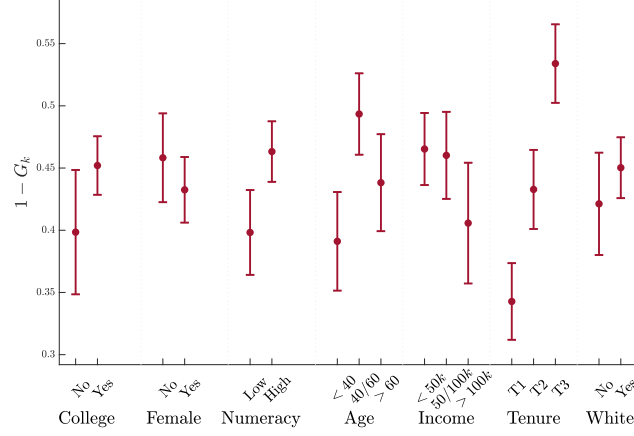
Table A.8: Belief stickiness and uncertainty

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$
$\ln(\sigma_{j,t+h,t-1})$	-0.261*** (0.045)	-0.262*** (0.045)	-0.262*** (0.045)
$\ln(\hat{\sigma}_{e,j,t})$	0.355*** (0.028)	0.354*** (0.028)	0.354*** (0.028)
$\pi_{t,t-12}$	0.004 (0.009)		
$\pi_{t,t-1}$		-0.011 (0.040)	
$\pi_{t-1,t-2}$			-0.004 (0.038)
Year-Month FEs	N	N	N
Group FE	Y	Y	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.63	0.63	0.63
Observations	768	768	768

Notes: Standard errors (in parentheses) are bootstrapped within and across group-month (j, t) level. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

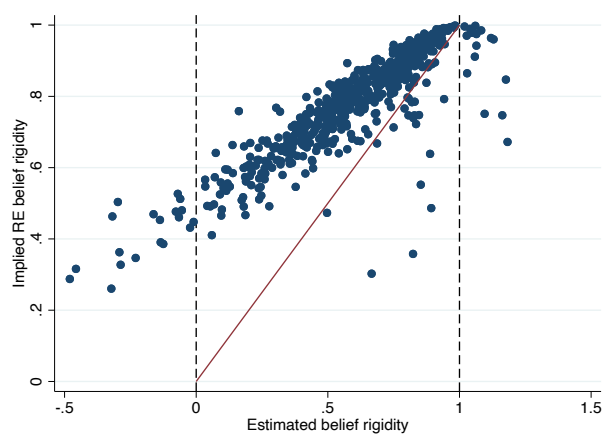
F Additional figures

Figure A.2: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: only households revising their answer from previous month



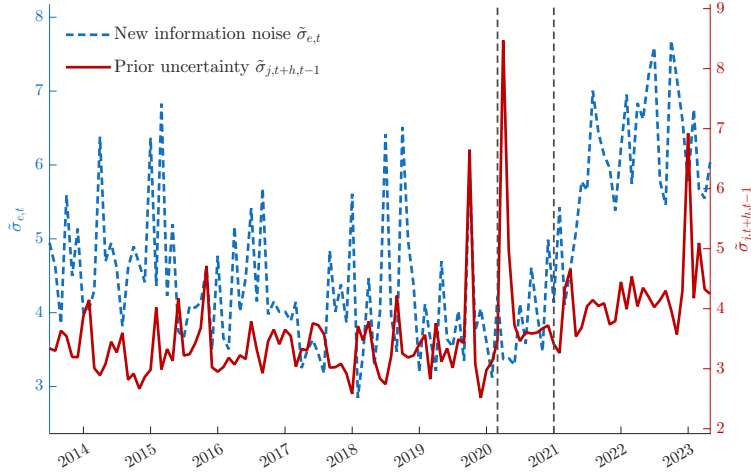
Notes: The figure reports the group-specific belief stickiness from (7), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. This figure only considers households providing in t a different forecast than in the previous month. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.3: Belief stickiness and uncertainty: consumer with large tenure



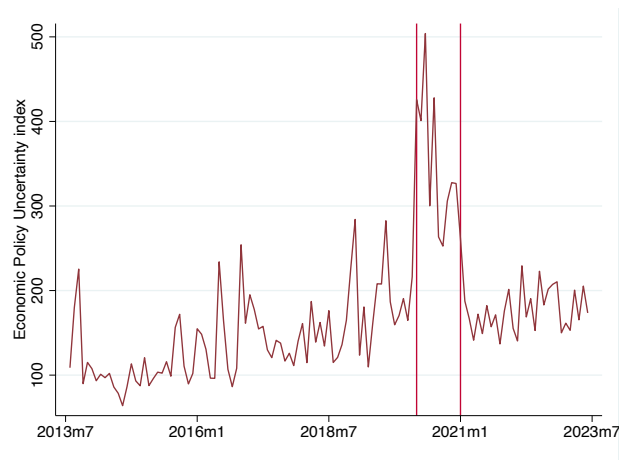
Notes: The figure plots on the x-axis the estimated belief stickiness $\hat{\beta}_{j,t}$ from regression (6) against the implied rational expectation belief stickiness from equation (8) on the y-axis. The sample excludes consumers in the first quartile of the tenure distribution.

Figure A.4: Decomposing uncertainty: prior and new information noise under Rational Expectation



Notes: The figure plots the estimates of new information noise (left y-axis) and prior uncertainty (right y-axis) from equation (14), with the estimates of estimates of belief stickiness $1 - \widehat{G}_t$, posterior uncertainty $\hat{\sigma}_{j,t+h,t}$ and $\alpha = 0, \theta = 0$.

Figure A.5: Economic Policy Uncertainty index



Notes: The figure plots the "news coverage component" from the Economic Policy Uncertainty Index, which is based on the share of articles in US online newspapers that mention economic policy uncertainty (Baker et al., 2016). This index is based on the share of articles in 10 large newspapers (USA Today, the Miami Herald, the Chicago Tribune, the Washington Post, the Los Angeles Times, the Boston Globe, the San Francisco Chronicle, the Dallas Morning News, the Houston Chronicle, and the WSJ) containing the terms 'uncertainty' or 'uncertain', the terms 'economic' or 'economy' and one or more of the following terms: 'congress', 'legislation', 'white house', 'regulation', 'federal reserve', or 'deficit'. For additional details, see Baker et al. (2016).

G Belief stickiness and uncertainty: alternative empirical strategy

We investigate the relationship between uncertainty and belief stickiness with a different empirical strategy compared to section 3.1.2. Instead of regressing prior and posterior uncertainty on the estimated group-month belief stickiness, we estimate belief stickiness and its relationship with uncertainty in the same regression by using an interaction term. That is, we regress

$$E_{i,t}[x_{t+h}] = \alpha + \beta_1 E_{i,t-1}[x_{t+h}] + \mathbf{X}_{i,t} \mathbf{B}_2 + E_{i,t-1}[x_{t+h}] \times \mathbf{X}_{i,t} \mathbf{B}_3 + \gamma_t + \beta_4 (\gamma_t \times \mathbf{X}_{i,t}) + err_t^i \quad (\text{A.2})$$

where $\mathbf{X}_{i,t}$ is a vector containing the measure of prior and a proxy for new information noise, namely posterior uncertainty or absolute forecast error. Table A.9 shows that the results are consistent with the ones in Table 1 and with Bayesian updating: belief stickiness is higher if new information is noisy and lower if the prior uncertainty is higher.

Table A.9: Belief stickiness and uncertainty

	(1) for _{3y}	(2) for _{3y}	(3) for _{3y}	(4) for _{3y}
$E_{t-1}^i[x_{t+h}]$	0.523*** (0.024)	0.065** (0.032)	0.467*** (0.023)	0.157*** (0.023)
$E_{t-1}^i[x_{t+h}] \times \ln(\sigma_{i,t+h,t-1})$	-0.122*** (0.016)	-0.128*** (0.012)		
$High(\sigma_{i,t+h,t-1})=1 \times E_{t-1}^i[x_{t+h}]$			-0.188*** (0.023)	-0.126*** (0.027)
$High(\sigma_{i,t+h,t-1})=1 \times E_{t-1}^i[x_{t+h}]$				
$E_{t-1}^i[x_{t+h}] \times \ln(\sigma_{i,t+h,t})$	0.106*** (0.015)			
$E_{t-1}^i[x_{t+h}] \times \ln(abs(FE_{i,t}))$		0.250*** (0.013)		
$High(\sigma_{i,t+h,t})=1 \times E_{t-1}^i[x_{t+h}]$			0.241*** (0.019)	
$High(abs(FE_{i,t}))=1 \times E_{t-1}^i[x_{t+h}]$				0.488*** (0.015)
Constant	2.532*** (0.069)	3.420*** (0.089)	2.344*** (0.057)	2.795*** (0.052)
Year-Month FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Year-Month $\times$ variables	Y	Y	Y	Y
Socioeconomic controls	Y	Y	Y	Y
Sample	Jun13-Feb20	Jun13-Feb20	Jun13-Feb20	Jun13-Feb20
Adjusted R-squared	0.39	0.50	0.37	0.39
Observations	55706	62786	65192	62786

Notes: This table reports the estimated coefficient from regression (A.2). We control for time fixed effect and their interaction with the variables. Standard errors (in parentheses) are clustered at the group-month level. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

H Aggregate belief stickiness

We compute belief stickiness by running the following panel regression in a rolling window of three months

$$E_t^i[x_{t+h}] = \alpha + \beta E_{t-1}^i[x_{t+h}] + \gamma_t + X_{i,t} + err_t^i \quad (\text{A.3})$$

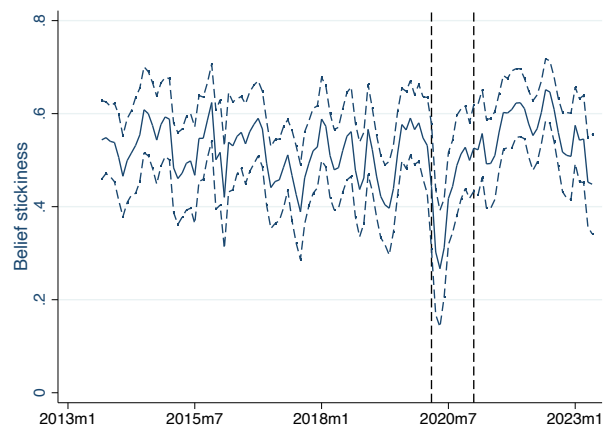
where $X_{i,t}$ contains sociodemographic controls, such as tenure (i.e. number of months in the survey), whether holding a college degree, age, income over 100k or below 50k, high numeracy, gender, and race. Figure A.6 plots the estimated belief stickiness $\hat{\beta}$ for each final month of the 3-month rolling window. Figure A.7 reports the same regression, but considering only the intensive margin of belief adjustment, meaning excluding observations where the posterior (forecast in t) equals the prior (forecast in $t - 1$), with the same result.

However, in the presence of heterogeneity in belief updating across households, as captured in our model (3), estimating aggregate belief stickiness through (A.3) is biased. To see this, we demean posterior expectations in (3) by the aggregate (rather than group-specific) and get:

$$\begin{aligned} E_{i,j,t}[x_{t+h}] - \int^i E_{i,j,t}[x_{t+h}] di = \\ (1 - G_{j,t})E_{i,j,t-1}[x_{t+h}] - \int^i (1 - G_{j,t})E_{i,j,t-1}[x_{t+h}] di + (G_{j,t} - \int^i G_{j,t} di)x_{t+h} + G_{j,t}e_{i,j,t}, \end{aligned} \quad (\text{A.4})$$

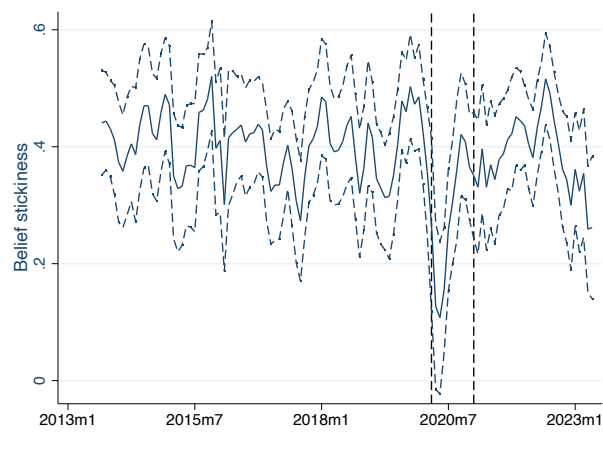
which implies that $\hat{\beta}$ in (A.3) does not map directly to the average belief stickiness $\overline{1 - G} = \int^i (1 - G_j) di$. For this reason, we rely on the group-level regression in equation (6). Nonetheless, the dynamics around the COVID-19 outbreak look similar in both approaches, reinforcing the robustness of our baseline results.

Figure A.6: Belief stickiness pre- and post-pandemic



Notes: The figure plots the estimate of [A.3](#) excluding observations where the posterior equals the prior, meaning excluding revisions equal to zero. The dashed blue lines represent the 95% confidence interval. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

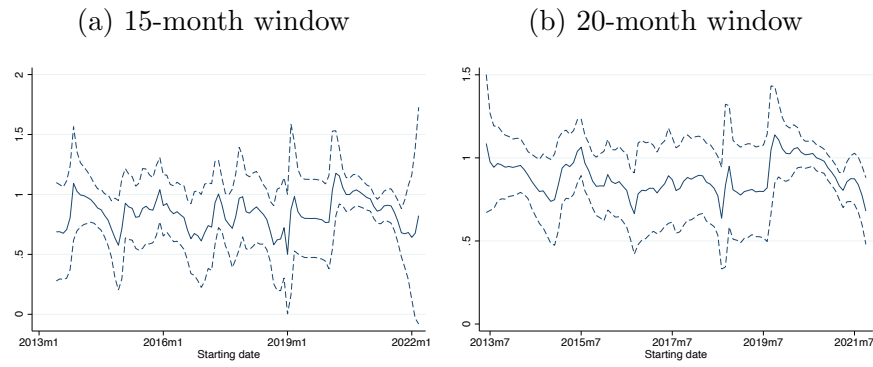
Figure A.7: Belief stickiness pre- and post-pandemic (non-zero revisions)



Notes: The figure plots the estimate of [A.3](#), while the dashed blue lines represent the 95% confidence interval. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

I Estimation of inflation autoregressive structure

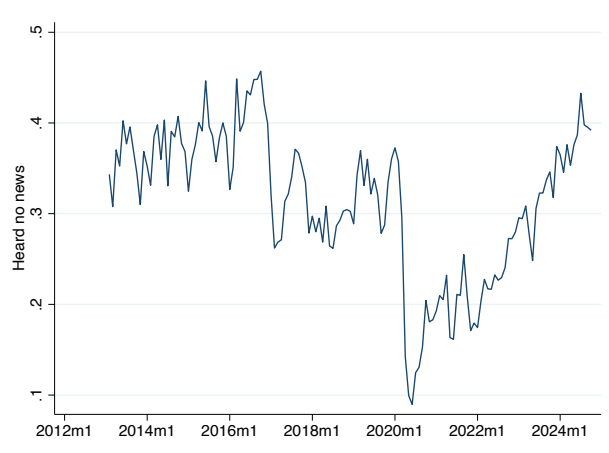
Figure A.8: Rolling window autoregressive estimation of monthly inflation



Notes: This figure show the estimates of an AR(1) autoregressive process for monthly inflation in a rolling window.

J Michigan Survey of Consumers

Figure A.9: Share of respondents who heard no news about Business Conditions



We consider a question in the Michigan Survey of Consumers: "During the last few months, have you heard of any favorable or unfavorable changes in business conditions?". In each month, we compute the share of respondents that answer "No". Figure A.9 reports this share, which declines at the onset of the COVID-19 pandemic, only to increase back in the following periods. While we can distinguish whether this is driven by demand (consumer) or supply (media) factors, it is nevertheless consistent with the evidence that consumers look for new information during the COVID period and less afterwards.

K Lockdown Impact during Covid

Measuring Lockdown Stringency. We measure the US state-level stringency of lockdown policies from the Oxford COVID-19 Government Response Tracker (OxCGRT) database. The database covers the period between January 2020 and December 2022 and contains information about closure and containment restrictions, which are recorded as ordinal categorical scales measuring the intensity or severity of the policy. Details about the collection process for a variety of countries are in [Hale et al. \(2020\)](#), while [Hallas et al. \(2021\)](#) provides an overview of the policy implemented at the US state level. We consider the following indicators: *school closing*, *workplace closing*, *cancel public events*, *restrictions on gathering size*, *close public transport*, *stay at home requirements*, and *restrictions on internal movements*. As the severity of these policies differs between vaccinated and non-vaccinated individuals, we consider the state average weighted by the number of vaccinated and non-vaccinated individuals. Finally, we compute a summary measure of the severity of lockdown measures, *lockdown*, equal to the simple average of these indicators.³¹ Figure 9 reports the time series of the country-level average of each indicator. Moreover, to measure the local impact of the pandemic we use the US state-level monthly level of COVID deaths and cases per capita. Table A.4 reports the summary statistics.

Figure 9 reports the estimated impact of lockdown indexes on belief stickiness, β_2 , while Table A.5 reports the detailed result. While all the indicators have a robust and negative effect on belief stickiness, including all of them together might create collinearity issues. As a result, we use the average of the indeces as a summary of the individual indicators. Once again the impact on belief stickiness is negative and robust. This result suggests that lockdown policies might have lowered the cost of collecting information for consumers, leading them to adjust their beliefs more than before.

Table A.10 presents additional evidence. The first column replicates the last column of Table A.5, using the average index *Lockdown* to summarize the stringency of state-level lockdown policies. The impact of lockdown policies on belief stickiness is still negative and robust. In the next three columns, we compare the effect of lockdown policies with measures of state-level economic policy uncertainty, from [Baker et al. \(2022\)](#). The indices are constructed from articles in local newspapers containing terms such as ‘economic’ and

³¹This measure is similar to the *stringency index* in [Hale et al. \(2020\)](#), as they also consider a simple average of each indicator. However, differently from them, we exclude from this average the indicators on *restrictions on international travel*, as not related to state-level measures, and *public information campaign*, as not related to lockdown measures.

Table A.10: Belief stickiness and lockdown measures

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.600*** (0.113)	0.604*** (0.116)	0.606*** (0.119)	0.603*** (0.118)
$E_{i,t-1}[x_{t+h}] \times \text{Lockdown}$	-0.082*** (0.030)	-0.083*** (0.030)	-0.083*** (0.030)	-0.082*** (0.030)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{DeathsCOVID})$	-0.017 (0.016)	-0.016 (0.016)	-0.016 (0.016)	-0.016 (0.016)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{CasesCOVID})$	0.032* (0.016)	0.032* (0.016)	0.032* (0.016)	0.032* (0.016)
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU State})$		0.010 (0.014)		
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU National})$			0.005 (0.016)	
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU Composite})$				0.007 (0.021)
Constant	2.451*** (0.085)	2.451*** (0.085)	2.453*** (0.085)	2.452*** (0.085)
Year-Month FEs	Y	Y	Y	Y
Socio-demographic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.35	0.35	0.35	0.35
Observations	25398	25397	25381	25398

Notes: E_t denotes the 1-year ahead forecast of inflation expectations starting 24 months into the future from the NY Fed Survey of Consumer Expectations (SCE). $Prior$ is the point forecast about the 3-year horizon provided in the previous month. $DeathsCOVID$ and $CasesCOVID$ are respectively the state-level COVID-related deaths and cases per capita. The EPU_{state} , $National$, and $Composite$ are the state-level economic policy uncertainty indicators from Baker et al. (2022). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

‘uncertainty’, and are divided according to the topic of the economic policy considered: national-level, state-level, and a composite of the two.³² Even controlling for state-level uncertainty, the estimated impact of lockdown policies on belief stickiness is significant and negative.³³

Lower information-gathering costs due to lockdown policies can explain the decrease in belief stickiness observed at the pandemic’s onset. However, is it also consistent with the sharp increase in belief uncertainty in the same period? We investigate this question in the following Section.

³²We take the percentage change in the measure to isolate the surprise component. The results are robust to using simple differences and levels.

³³Tables A.13 reports the results for the one year inflation forecasts, with similar results.

L Belief uncertainty and disagreement

In this section we clarify the difference between belief uncertainty, namely the variance of posterior beliefs, and disagreement, meaning the cross-sectional dispersion in posterior mean. We show that new information noise unambiguously increases the former, but not the latter.

Simple model Consider a simple version of the belief-updating setting considered in section 3 where the variable forecasted is i.i.d. Suppose agents form beliefs about stochastic variable $x \sim N(\mu_x, \sigma_x^2)$ where μ_x is the prior mean and σ_x^2 is the prior variance. Agents can not observe x directly, but receive a private noisy signal about it, similarly to equation (2)

$$s^i = x + e^i \quad (\text{A.5})$$

where the signal noise $e^i = \eta^i + \omega$ contains (i) an idiosyncratic component η^i normally distributed mean-zero noise with variance σ_η^2 and i.i.d. across time and households, i.e. $\int^i e_t^i di = 0$, and (ii) a common component ω normally distributed mean-zero noise with variance σ_ω^2 which is i.i.d. only across time, but not across agents. Let $\sigma_e^2 \equiv \sigma_\eta^2 + \sigma_\omega^2$ define the overall variance of the signal noise.

The Bayesian posterior beliefs is $x|s^i \sim N(E^i[x|s^i], \text{Var}^i[x|s^i])$. The posterior mean equals

$$E^i[x|s^i] = (1 - G)\mu_x + Gs^i \quad (\text{A.6})$$

where the Bayesian weight on new information is $G = \frac{\sigma_x^2}{\sigma_e^2 + \sigma_x^2} = \frac{\sigma_x^2}{\sigma_\eta^2 + \sigma_\omega^2 + \sigma_x^2}$.

Belief uncertainty The Bayesian posterior variance, or uncertainty, then equals

$$\text{Var}^i[x|s^i] = \sigma_e^2 G = \frac{\sigma_x^2 \sigma_e^2}{\sigma_e^2 + \sigma_x^2} \quad (\text{A.7})$$

Therefore

$$\frac{\partial \text{Var}^i[x|s^i]}{\partial \sigma_e^2} = \left(\frac{\sigma_x^2}{\sigma_e^2 + \sigma_x^2} \right)^2 > 0 \quad (\text{A.8})$$

Posterior uncertainty increases in the new information noise, no matter whether the increase is due to new private information noise σ_η^2 or new public information noise σ_ω^2 . In section 3, we do not need to take a stand about what drives the increase in new information noise to derive our main implications.

Belief disagreement Let's consider now disagreement, meaning cross-sectional dispersion of posterior mean across agents.

$$Disp(E^i[x|s^i]) = G^2 \sigma_\eta^2 \quad (\text{A.9})$$

As we consider the second moment of the cross-sectional distribution, the common error and realization across forecasters drop out.

First, consider an increase in public information noise σ_ω^2 . Intuitively, there is no direct effect on disagreement, as new information received by agents becomes equally more uncertain. If all information were public, then there would be no effect at all on disagreement. However, since the new signal also contains private information, an increase in public noise has an indirect effect on belief dispersion: as the new signal is overall noisier, agents allocate less weight G to it, and more to the common prior, leading to a decrease in disagreement:

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\omega^2} = \sigma_\eta^2 \frac{\partial G^2}{\partial \sigma_\omega^2} = -2 \frac{\sigma_x^2 \sigma_\eta^2}{(\sigma_\eta^2 + \sigma_\omega^2 + \sigma_x^2)^2} < 0 \quad (\text{A.10})$$

Now consider an increase in private information noise σ_η^2 . In this case, there are two effects. First, a direct effect: larger volatility of idiosyncratic shocks makes new information more dispersed across agents. This is represented by the first term on the right-hand side of equation (A.11). Second, an indirect effect: as new information is overall noisier, agents allocate less weight to G to it, and more to the common prior. This is represented by the second term on the right-hand side of equation (A.11)

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\eta^2} = G^2 + \sigma_\eta^2 \frac{\partial G^2}{\partial \sigma_\eta^2} \quad (\text{A.11})$$

Therefore,

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\eta^2} > 0 \iff \sigma_\eta^2 < \frac{\sigma_x^2}{2} \quad (\text{A.12})$$

The first effect prevails for low values of σ_η^2 , while the second prevails for higher values of σ_η^2 . In other words, the effect of new private information noise on belief dispersion is non-monotonic.

To sum up, while an increase in new information noise unambiguously increases posterior uncertainty, the effect on belief disagreement is nuanced and can go in either direction.

M Belief stickiness and forecast errors

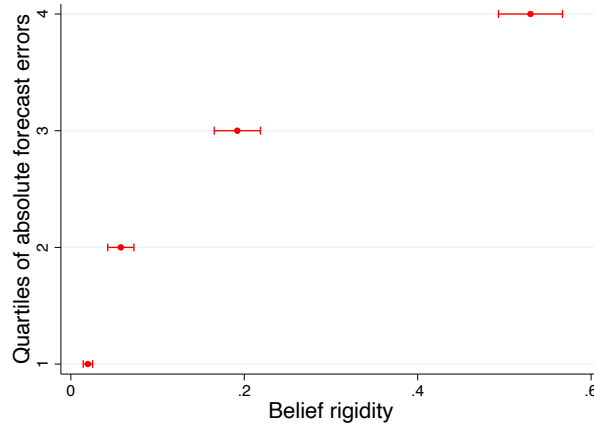
We investigate whether higher belief stickiness is associated with higher forecast errors. We define forecast error as the difference between the realization of 12-month inflation starting 24 months from the survey data and the individual forecast. We run the following regression

$$E_t^i[x_{t+h}] = \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \mathbf{FE}_{i,t} \mathbf{B}_2 + E_{t-1}^i[x_{t+h}] \times \mathbf{FE}_{i,t} \mathbf{B}_3 + \gamma_t + \beta_4(\gamma_t \times \mathbf{FE}_{i,t}) + err_t^i \quad (\text{A.13})$$

where $FE_{i,t}$ contains the dummy indicators with value 1 if the absolute forecast error of individual i at time t is in the first, second, third, or fourth quartile of the unconditional absolute forecast error distribution. The coefficient $\mathbf{B}_3$ identifies belief stickiness at different forecast error quartiles.

Figure A.10 reports the results. We find that the relation between belief stickiness and forecast errors is positive: higher belief stickiness is associated with higher forecast errors. Table A.11 reports alternative specifications. The results are robust to using absolute forecast errors and limiting the sample to pre-Covid period.

Figure A.10: Forecast errors



Notes: The figure shows the relationship between forecast errors and belief stickiness, $\mathbf{B}_3$ from regression A.13 as reported in column (3) of table A.11. Bars represent the 95% confidence interval. Sample period: 2013M6-2019M11.

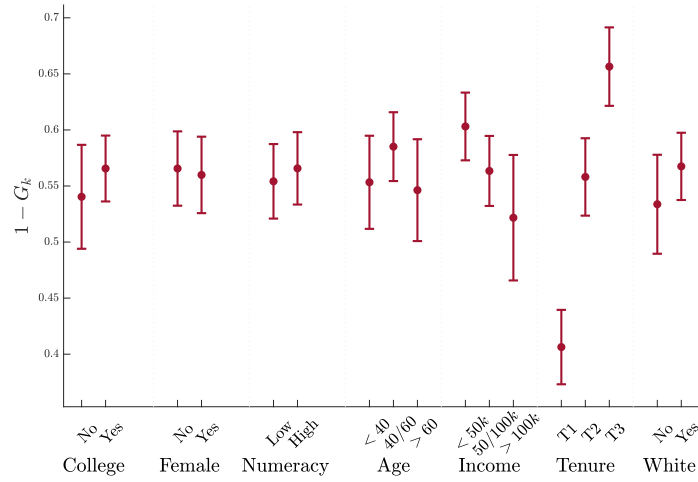
Table A.11: Forecast errors and belief stickiness

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.566*** (0.012)	0.180*** (0.013)	0.020*** (0.003)
$E_{i,t-1}[x_{t+h}] \times \text{abs}(FE)$		0.008*** (0.001)	
<i>Quartile abs(FE)=2</i> $\times E_{i,t-1}[x_{t+h}]$			0.058*** (0.008)
<i>Quartile abs(FE)=3</i> $\times E_{i,t-1}[x_{t+h}]$			0.192*** (0.014)
<i>Quartile abs(FE)=4</i> $\times E_{i,t-1}[x_{t+h}]$			0.530*** (0.019)
Constant	1.981*** (0.057)	3.530*** (0.087)	3.243*** (0.044)
Year-Month FEs	Y	Y	Y
Year-Month FEs $\times$ FE	N	Y	Y
Non-interacted FE	Y	Y	Y
Sample	Jun13-Dec19	Jun13-Dec19	Jun13-Dec19
Adjusted R-squared	0.32	0.59	0.43
Observations	63509	63509	63509

Notes: $\text{abs}(FE)$ indicates the absolute forecast error of the individual, defined as the absolute difference between the 12-month inflation starting 24 months into the future and *Forecast*. *Quartile abs(FE)* equals 1 if $\text{abs}(FE)$ lies in the specific quartile of the unconditional distribution of $\text{abs}(FE)$. The estimation period is 2013M6-2023M5. Standard errors (in parentheses) are clustered at the individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

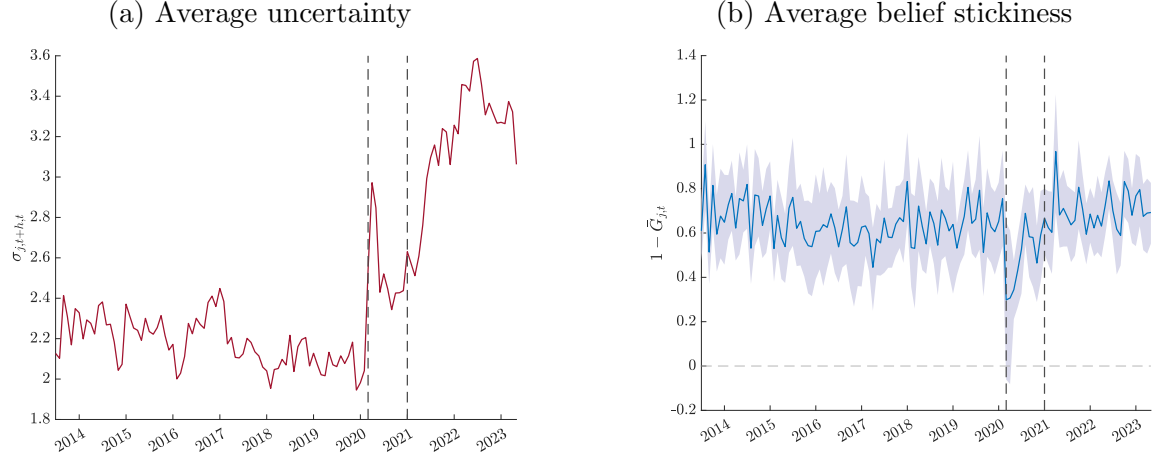
N Shorter forecast horizon

Figure A.11: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: 1-year horizon



Notes: The figure reports the group-specific belief stickiness from (7), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Expectations are about the 12-month ahead (1-year) inflation. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.12: Belief stickiness pre- and post-pandemic: 1-year horizon



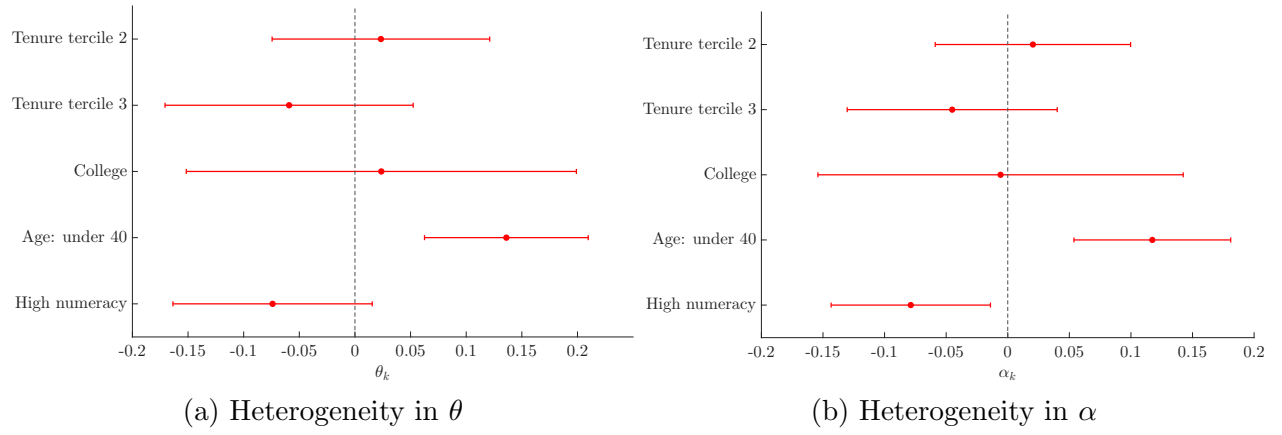
Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 6, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across (j, t) groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Table A.12: Belief stickiness and uncertainty: 1-year horizon

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$	(4) $1 - G_{j,t}$
$\ln(\sigma_{j,t+h,t-1})$	-0.206*** (0.034)		-0.392*** (0.029)	-0.249*** (0.045)
$\ln(\widehat{\sigma}_{e,j,t})$		0.281*** (0.021)	0.334*** (0.027)	0.324*** (0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	738	738	738	738

Notes: This table reports the estimated coefficient from regression (11). Standard errors (in parentheses) are computed with a two-level bootstrap. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Figure A.13: Heterogeneity in overreaction bias: 1-year horizon



Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 13. The right panel plots the estimated interactions with socioeconomic dummies and α in 13. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across (j, t) groups. Sample Jun13-Feb20.

Table A.13: Belief stickiness and lockdown measures: : 1-year horizon

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.788*** (0.213)	0.766*** (0.190)	0.789*** (0.206)	0.777*** (0.194)
$E_{i,t-1}[x_{t+h}] \times \text{Lockdown}$	-0.109*** (0.027)	-0.107*** (0.026)	-0.109*** (0.027)	-0.108*** (0.026)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{DeathsCOVID})$	0.008 (0.021)	0.007 (0.019)	0.008 (0.020)	0.008 (0.019)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{CasesCOVID})$	0.017 (0.016)	0.015 (0.016)	0.017 (0.016)	0.016 (0.016)
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU State})$		-0.020 (0.025)		
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU National})$			0.004 (0.020)	
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU Composite})$				-0.012 (0.030)
Constant	3.099*** (0.121)	3.103*** (0.121)	3.100*** (0.122)	3.102*** (0.122)
Year-Month FEs	Y	Y	Y	Y
Socio-demographic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.40	0.40	0.40	0.40
Observations	25263	25261	25248	25263

Notes: *DeathsCOVID* and *CasesCOVID* are respectively the state-level COVID-related deaths and cases per capita. The *EPUstate*, *National*, and *Composite* are the state-level economic policy uncertainty indicators from [Baker et al. \(2022\)](#). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Table A.14: Belief stickiness and lockdown measures: 1-year horizon

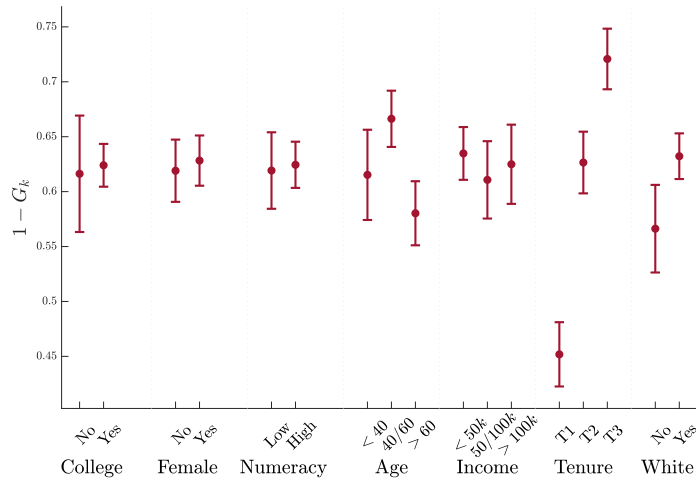
	(1)	(2)	(3)
	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$
<i>Lockdown</i>	-0.468*** (0.085)	-0.459*** (0.083)	-0.320*** (0.064)
$\sigma_{k,t+h-1,t-1}$			0.415*** (0.044)
$\ln(DeathsCOVID)$			0.011 (0.043)
$\ln(CasesCOVID)$			-0.046 (0.033)
$\ln(EPUNational)$			0.070* (0.037)
Constant	3.754*** (0.099)	3.748*** (0.053)	1.753*** (0.362)
State FEs	N	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0.07	0.29	0.42
Observations	1718	1718	1696

Notes: The table reports the results from regression 17. *EPUComposite* is the state-level economic policy uncertainty indicator from Baker et al. (2022). We control for state FEs. Covid controls are logged *DeathsCOVID* and *CasesCOVID*, which are the state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at State and time levels. *, **, and *** are 10%, 5%, and 1% confidence levels.

O Alternative measure of expected inflation: mean of subjective density forecasts

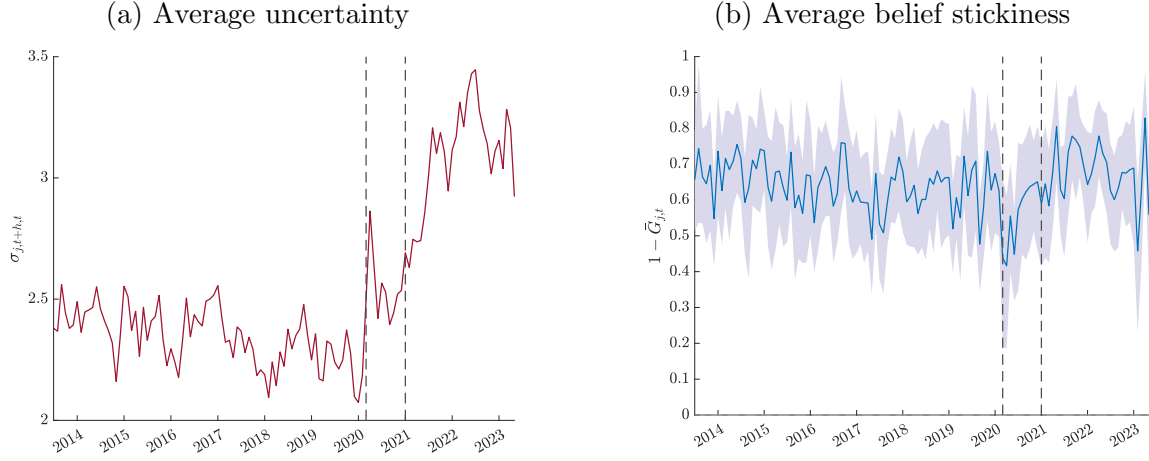
In the main text we use the point forecasts to measure expected mean inflation, i.e. the first moment of the subjective belief distribution, and the density forecasts uniquely to measure uncertainty, i.e. the second moment. Here, we instead measure the expected mean using the implied mean of the individual density forecasts.

Figure A.14: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: expectations from mean of density forecasts



Notes: The figure reports the group-specific belief stickiness from (7), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Expectations are not measured as the mean of the density forecast, as described in section. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.15: Belief stickiness pre- and post-pandemic: expectations from mean of density forecasts



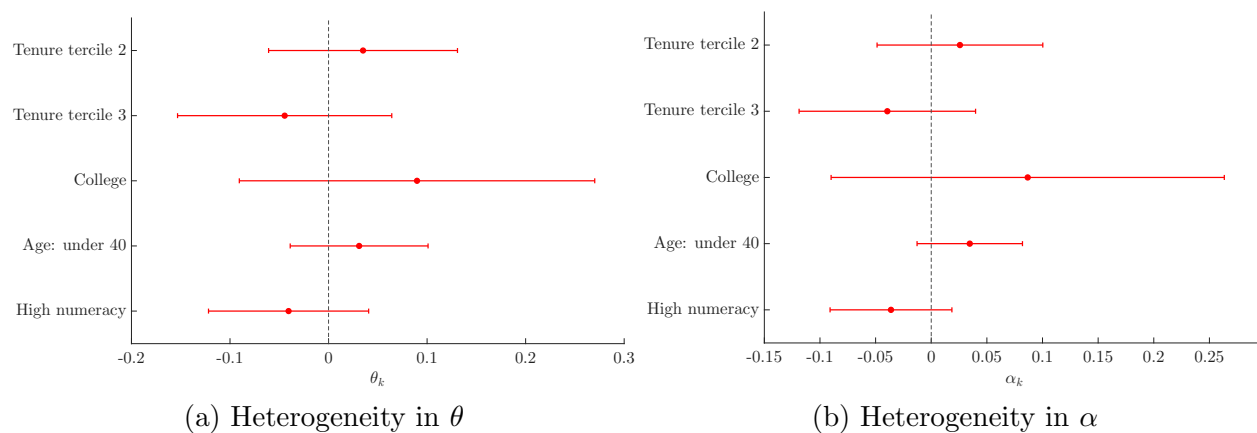
Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 6, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across (j, t) groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Table A.15: Belief stickiness and uncertainty: expectations from mean of density forecasts

	(1)	(2)	(3)	(4)
	$1 - G_{j,t}$	$1 - G_{j,t}$	$1 - G_{j,t}$	$1 - G_{j,t}$
$\ln(\sigma_{j,t+h,t-1})$	-0.206*** (0.034)		-0.392*** (0.029)	-0.249*** (0.045)
$\ln(\widehat{\sigma}_{e,j,t})$		0.281*** (0.021)	0.334*** (0.027)	0.324*** (0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	738	738	738	738

Notes: This table reports the estimated coefficient from regression (11). Standard errors (in parentheses) are computed with a two-level bootstrap. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Figure A.16: Heterogeneity in overreaction bias: expectations from mean of density forecasts



Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 13. The right panel plots the estimated interactions with socioeconomic dummies and α in 13. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across (j, t) groups. Sample Jun13-Feb20.

Table A.16: Belief stickiness and lockdown measures: expectations from mean of density forecasts

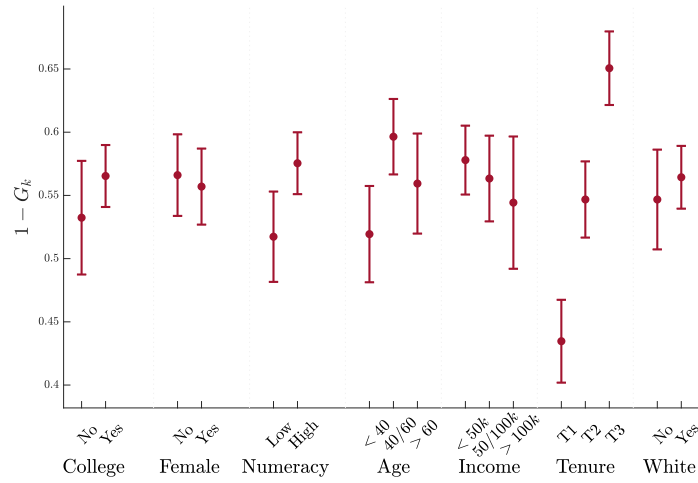
	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.938*** (0.130)	0.970*** (0.128)	0.955*** (0.135)	0.974*** (0.138)
$E_{i,t-1}[x_{t+h}] \times \text{Lockdown}$	-0.067** (0.028)	-0.068** (0.029)	-0.068** (0.028)	-0.070** (0.028)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{DeathsCOVID})$	0.019 (0.019)	0.017 (0.018)	0.019 (0.019)	0.021 (0.020)
$E_{i,t-1}[x_{t+h}] \times \ln(\text{CasesCOVID})$	0.020 (0.016)	0.028* (0.015)	0.021 (0.016)	0.022 (0.017)
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU State})$		0.029 (0.021)		
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU National})$			0.019 (0.017)	
$E_{i,t-1}[x_{t+h}] \times \Delta \ln(\text{EPU Composite})$				0.051** (0.021)
Constant	1.544*** (0.070)	1.536*** (0.065)	1.542*** (0.069)	1.542*** (0.069)
Year-Month FEs	Y	Y	Y	Y
Socio-demographic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.37	0.37	0.37	0.37
Observations	26007	26004	25990	26007

Notes: *DeathsCOVID* and *CasesCOVID* are respectively the state-level COVID-related deaths and cases per capita. The *EPUstate*, *National*, and *Composite* are the state-level economic policy uncertainty indicators from Baker et al. (2022). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at the individual and time levels. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

P Alternative calibration for prior forecast

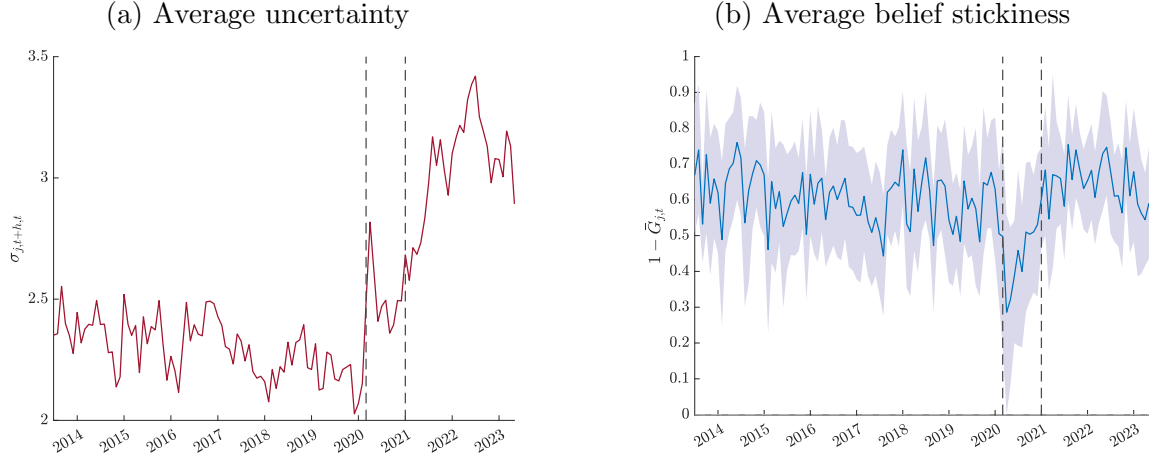
In the main text, we construct prior mean 3 and variance 9 using the estimated parameters $\rho = 0.946$, $\hat{\mu}_x = 2.3$ and $\sigma_u = 0.38$, reported in the first column of Table A.1. Here, we use an alternative calibration for this parameters, as reported in the second column of Table A.1: $\rho = 0.975$, $\mu_x = 2.82$ and $\sigma_u = 0.4$, with virtually identical results.

Figure A.17: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: alternative calibration for prior expectations



Notes: The figure reports the group-specific belief stickiness from (7), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.18: Belief stickiness pre- and post-pandemic: alternative calibration for prior expectations



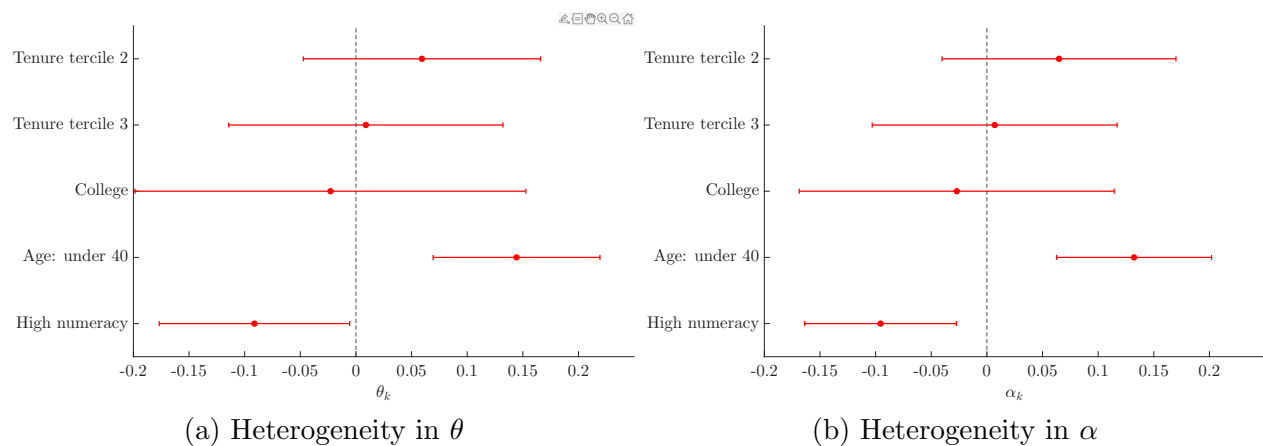
Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 6, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across (j, t) groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Table A.17: Belief stickiness and uncertainty: alternative calibration for prior expectations

	(1)	(2)	(3)	(4)
	$1 - G_{j,t}$	$1 - G_{j,t}$	$1 - G_{j,t}$	$1 - G_{j,t}$
$\ln(\sigma_{j,t+h,t-1})$	-0.206*** (0.034)		-0.392*** (0.029)	-0.249*** (0.045)
$\ln(\widehat{\sigma}_{e,j,t})$		0.281*** (0.021)	0.334*** (0.027)	0.324*** (0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	738	738	738	738

Notes: This table reports the estimated coefficient from regression (11). Standard errors (in parentheses) are computed with a two-level bootstrap. * represents $p < 0.10$, ** represents $p < 0.05$, and *** represents $p < 0.01$.

Figure A.19: Heterogeneity in overreaction bias: alternative calibration for prior expectations



Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 13. The right panel plots the estimated interactions with socioeconomic dummies and α in 13. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across (j, t) groups. Sample Jun13-Feb20.